

LAPORAN PRAKTIKUM
POSTTEST 7
ALGORITMA PEMROGRAMAN LANJUT

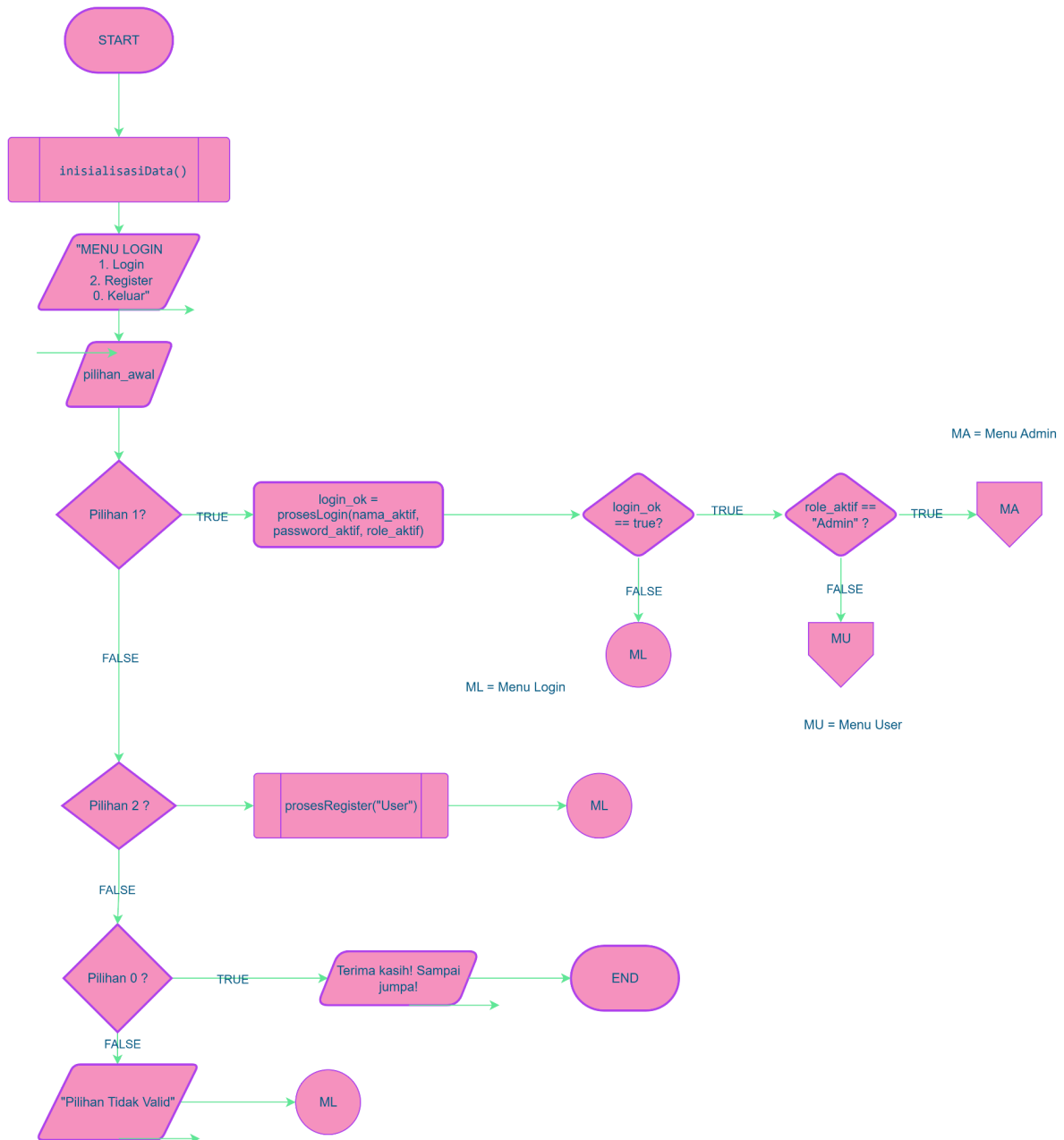


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A2 '25

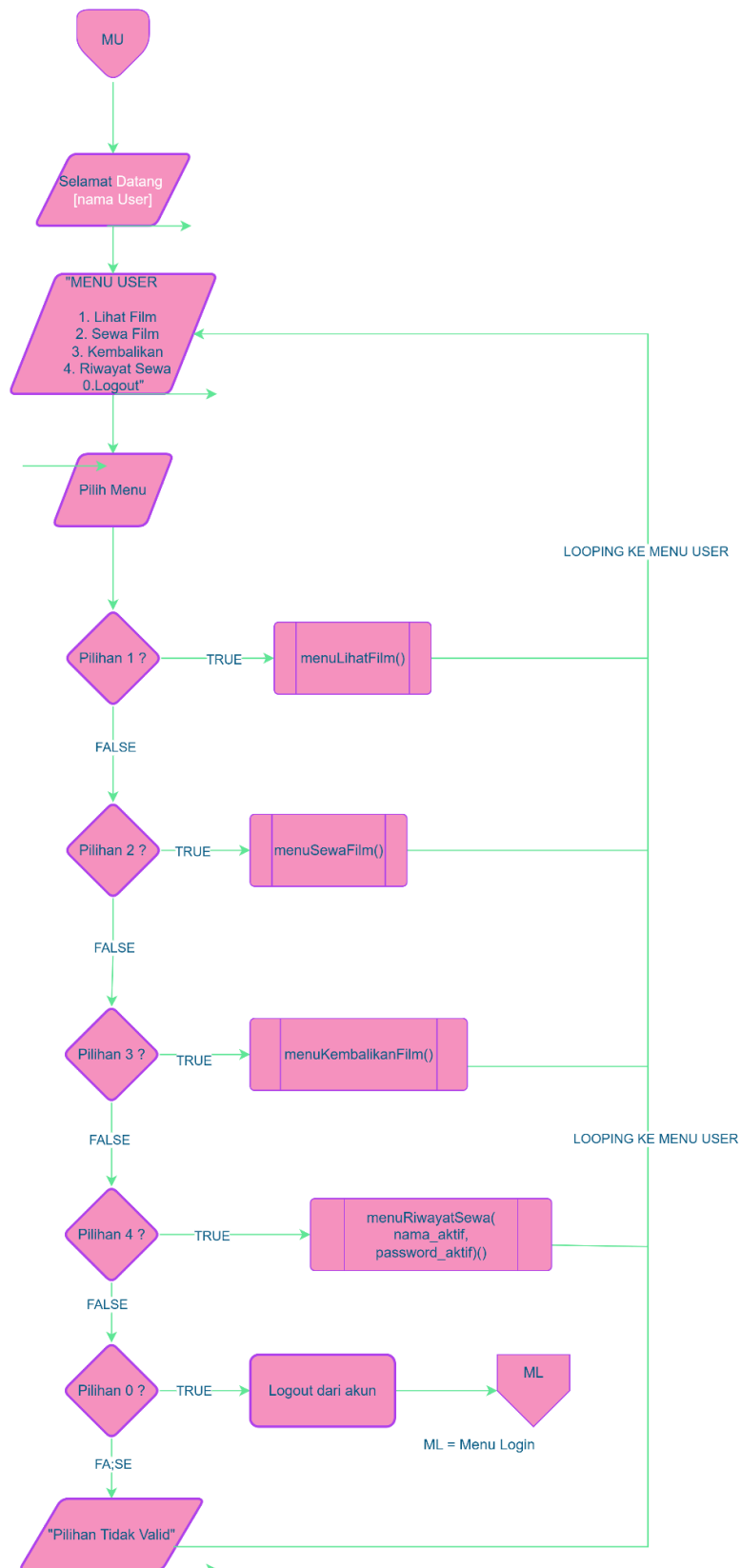
PROGRAM STUDI INFORMATIKA
UNIVERSITAS MULAWARMAN
SAMARINDA
2026

1. Flowchart

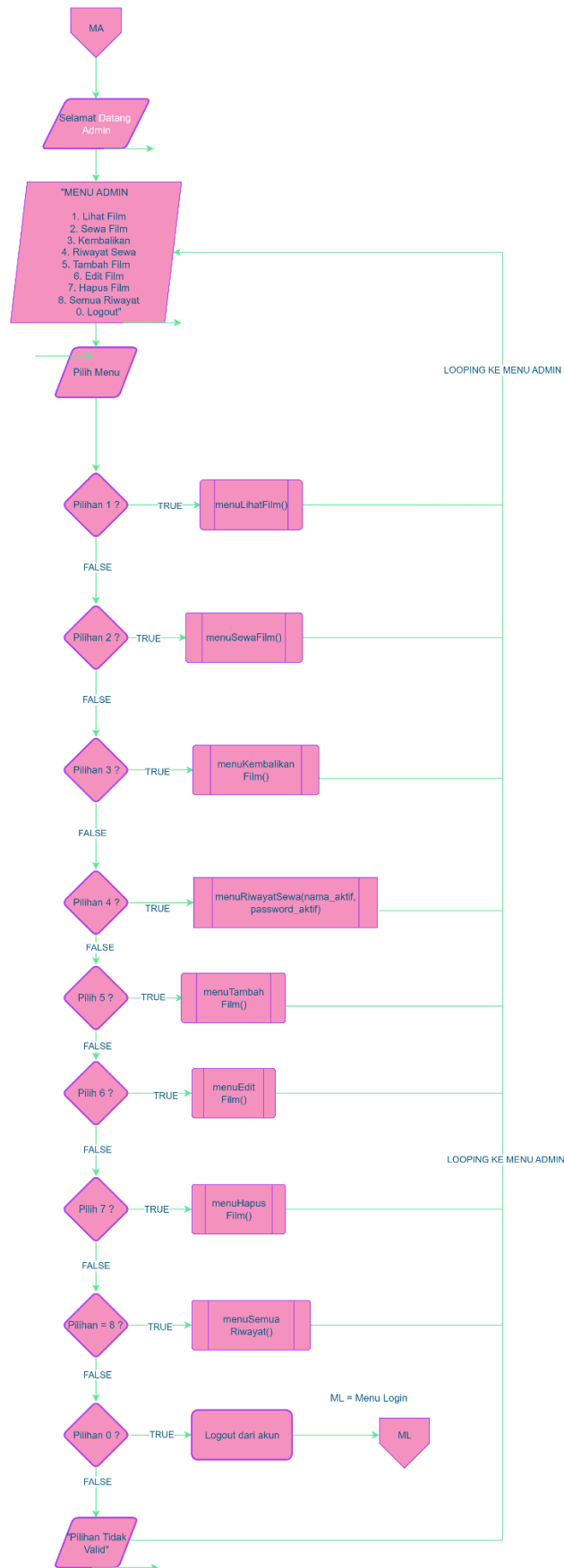
1.1 Gambar Bagian-Bagian Flowchart



Gambar 1.1 Flowchart Bagian Menu Login

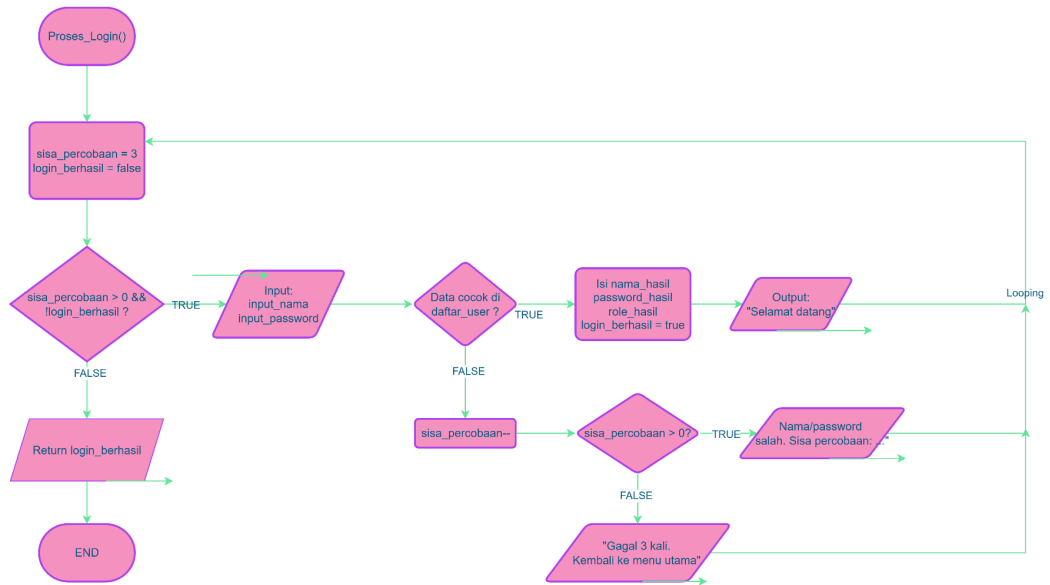


Gambar 1.2 Flowchart Bagian Menu User



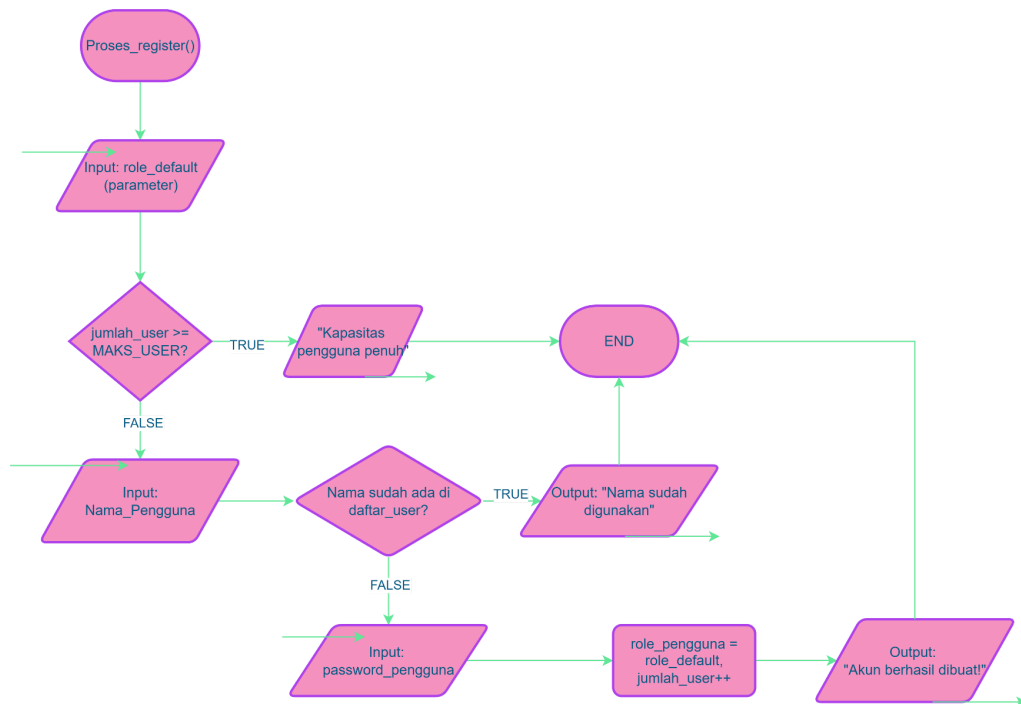
Gambar 1.3 Flowchart Bagian Menu Admin

PROSES
LOGIN

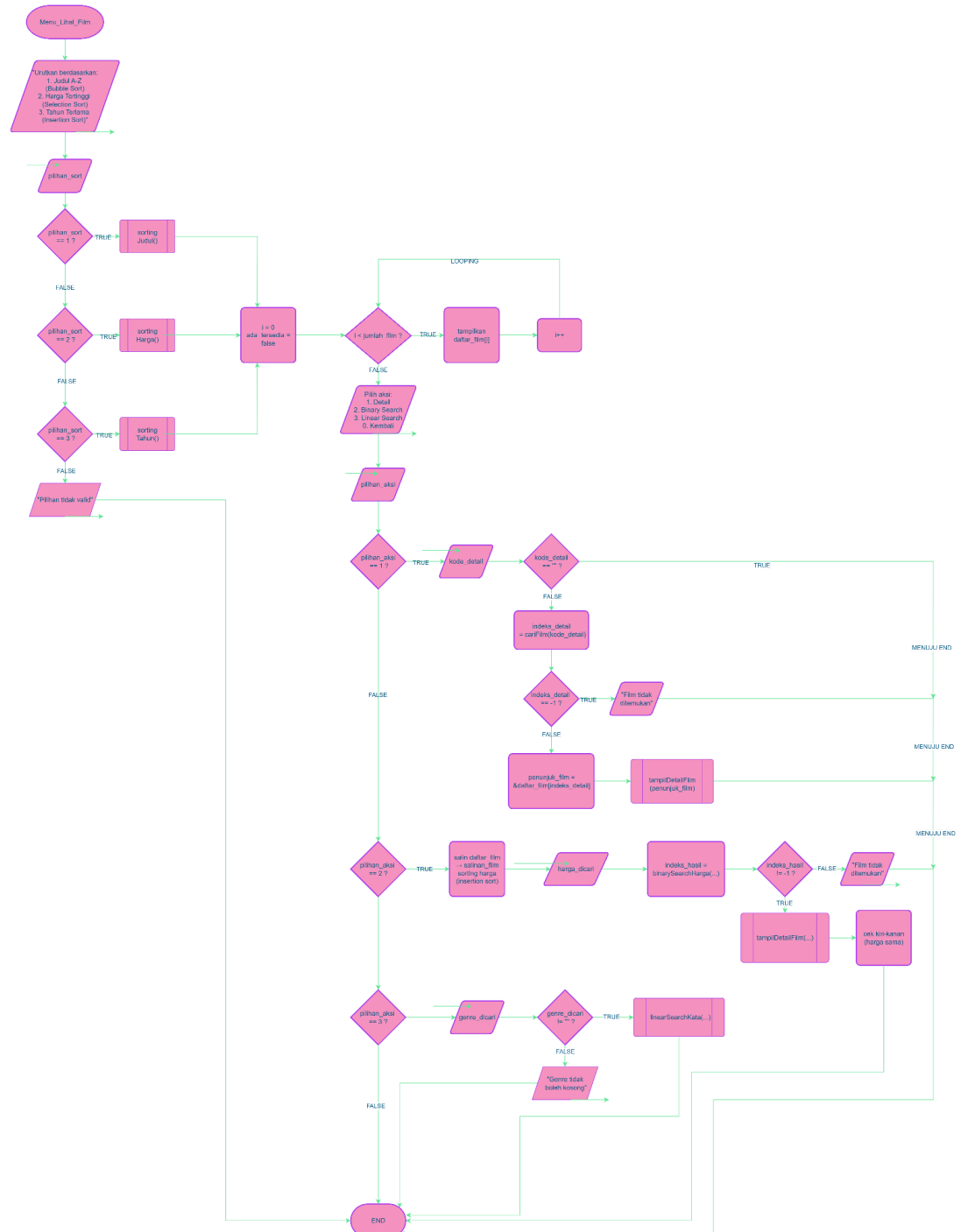


Gambar 1.4 Proses Login

PROSES
REGISTER

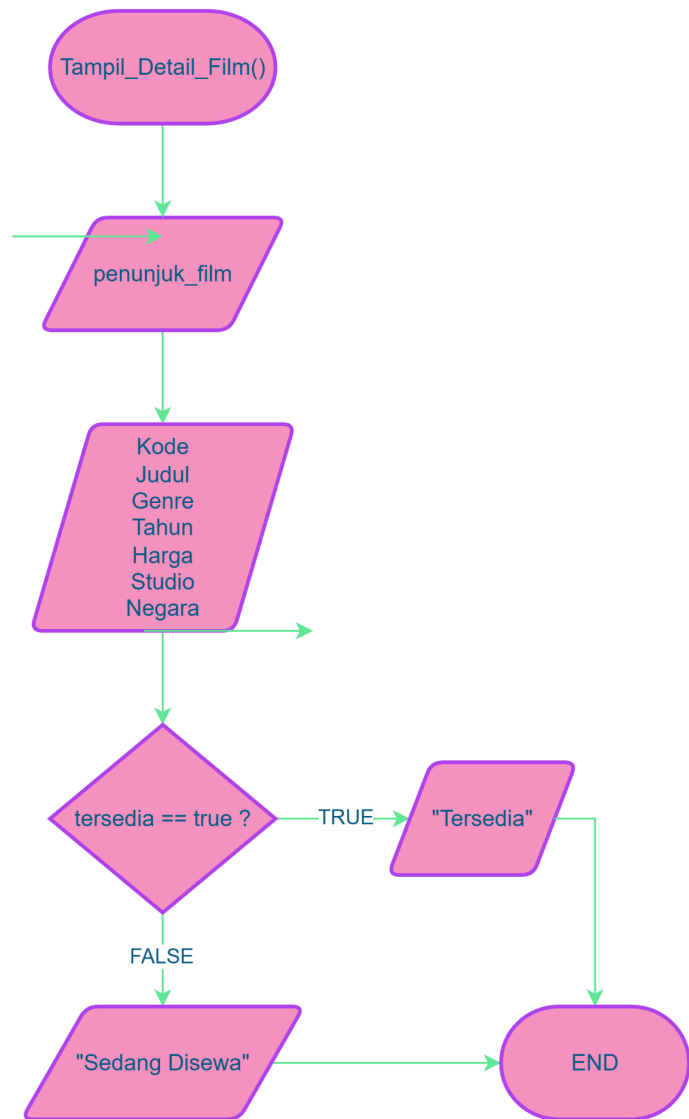


Gambar 1.5 Proses Register



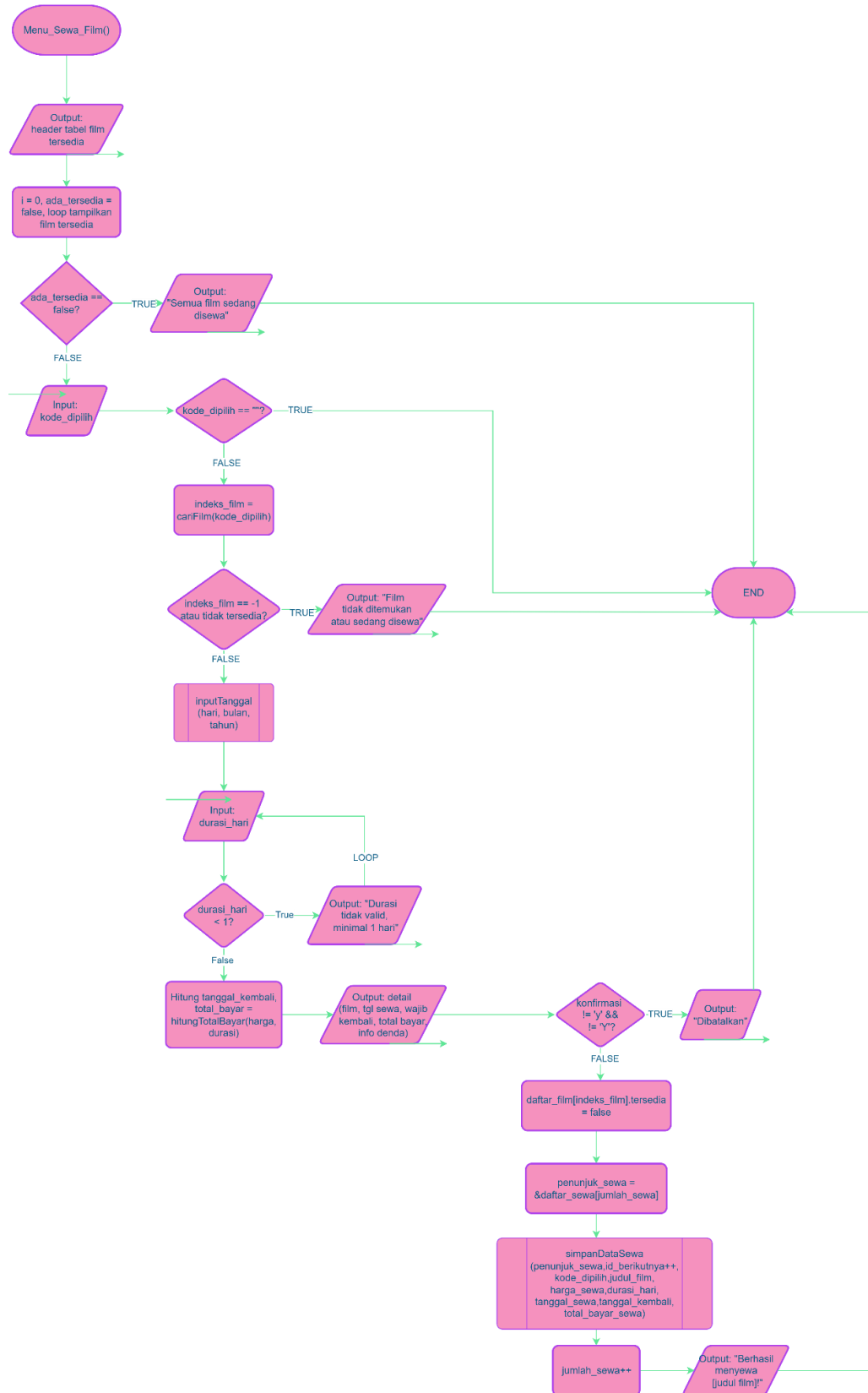
Gambar 1.6 Menu Lihat Film

TAMPIL DETAIL FILM

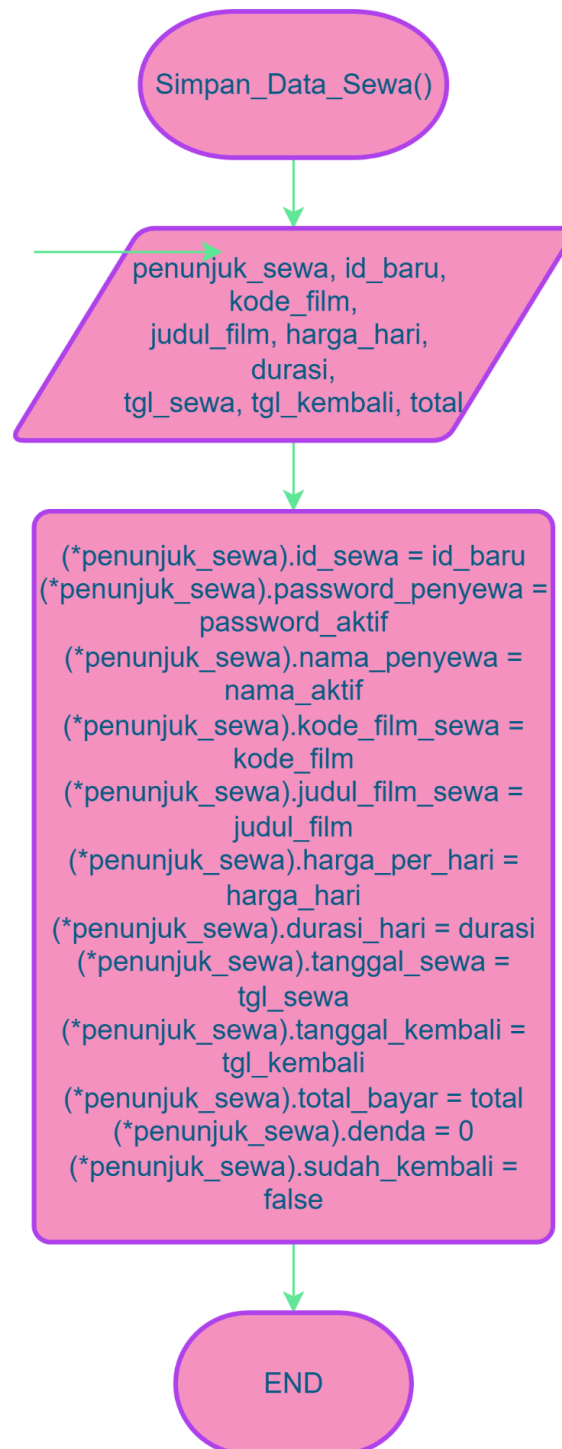


Gambar 1.7 Tampil Detail Film

menu sewa FILM

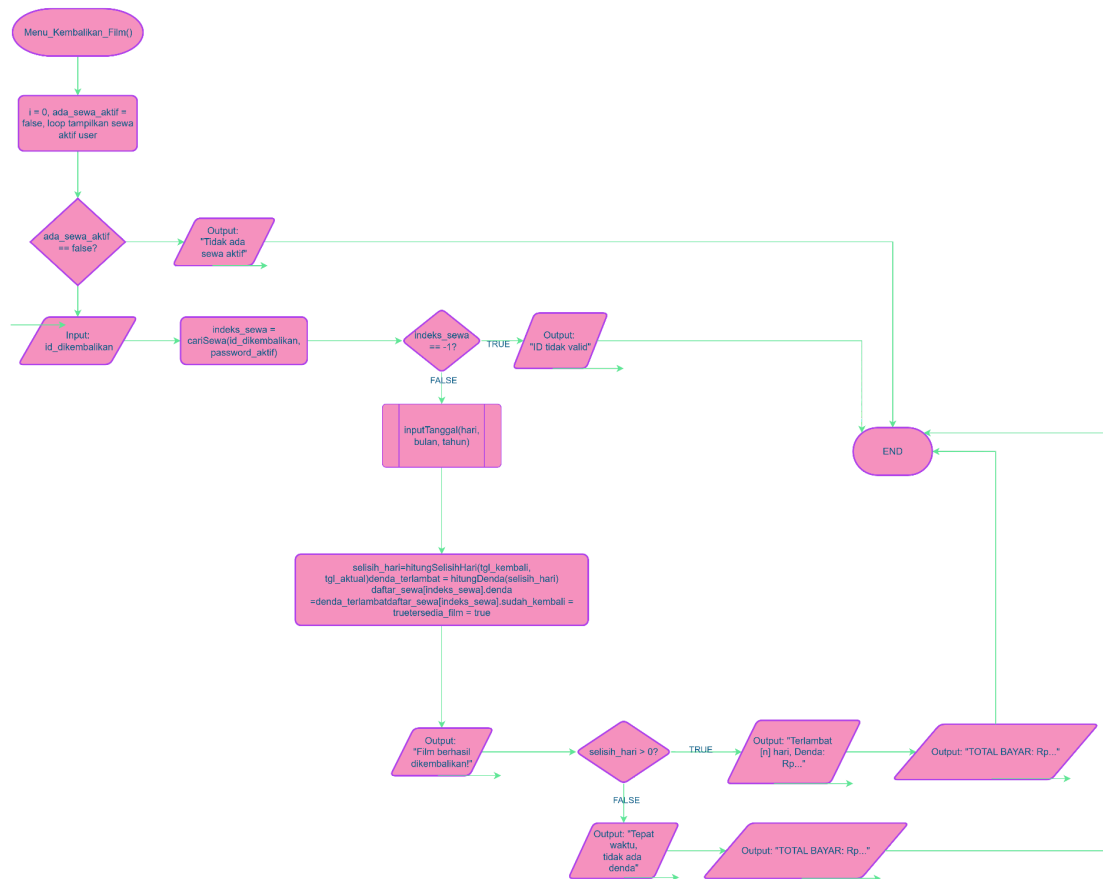


Gambar 1.8 Menu Sewa Film



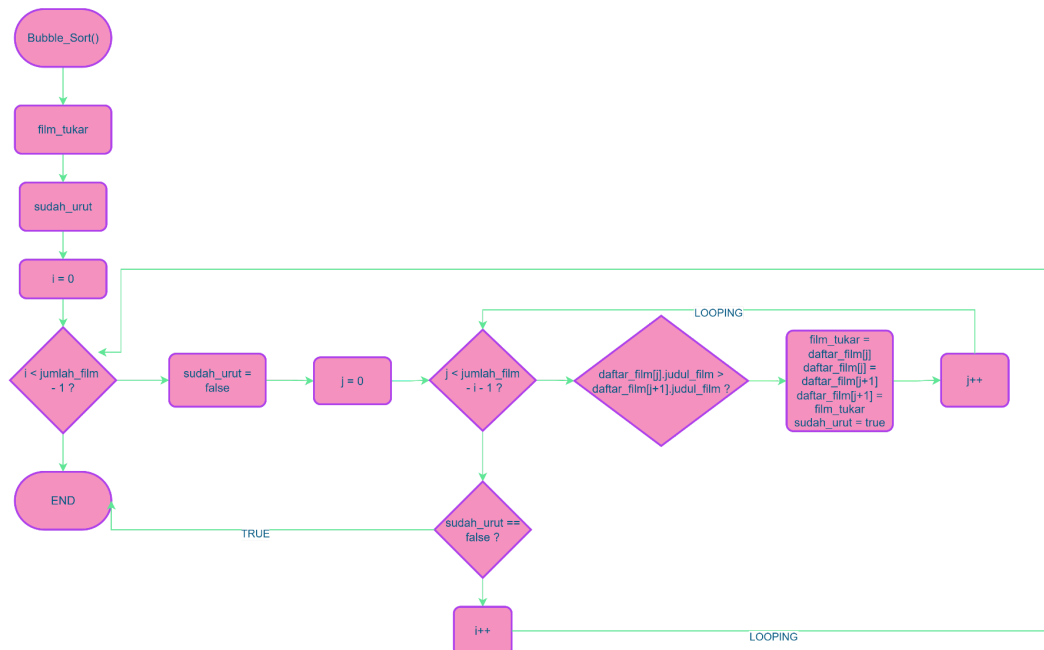
Gambar 1.9 Simpan Data Sewa

MENU
KEMBALIKAM
FILM

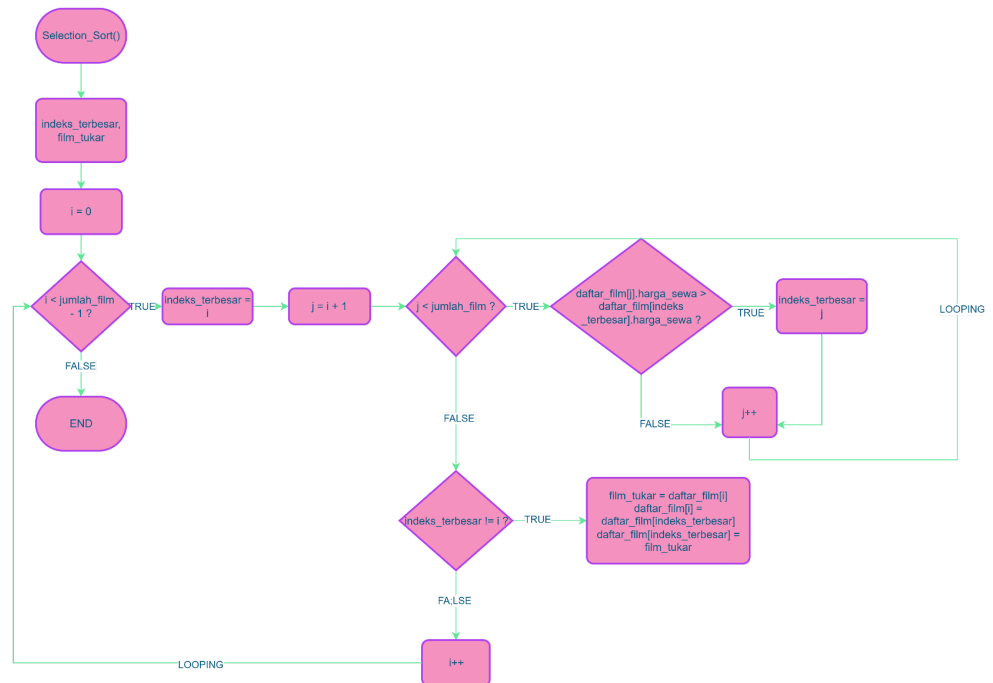


Gambar 1.10 Menu Kembalikan Film

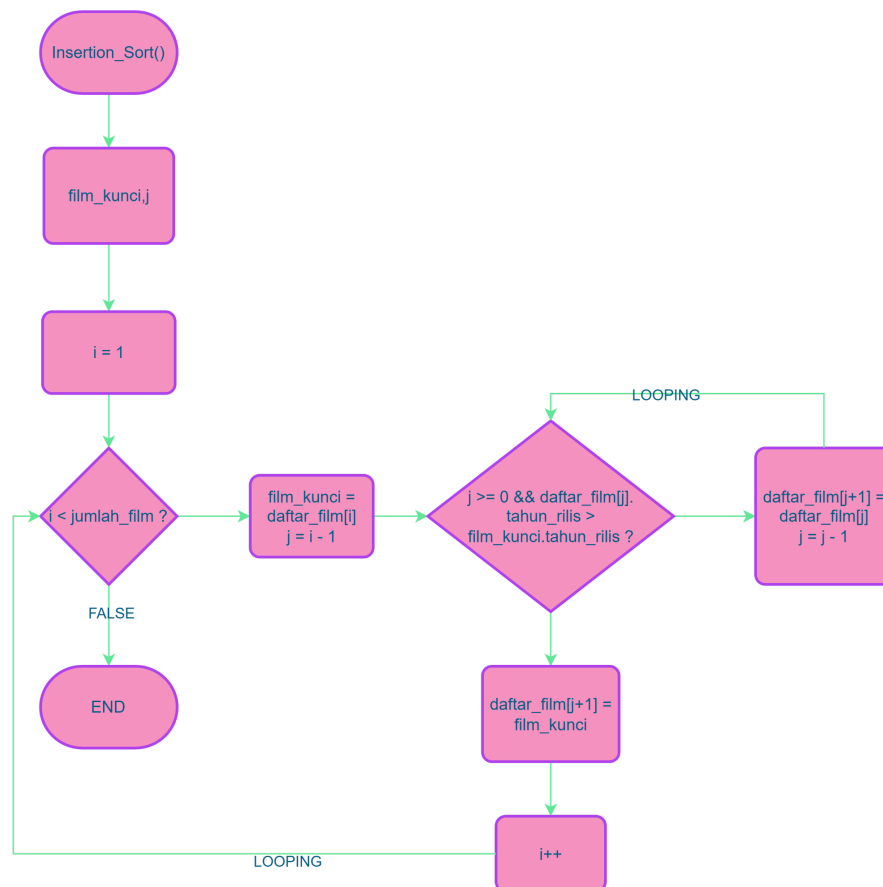
BUBBLE
SORT



Gambar 1.11 Sorting Judul

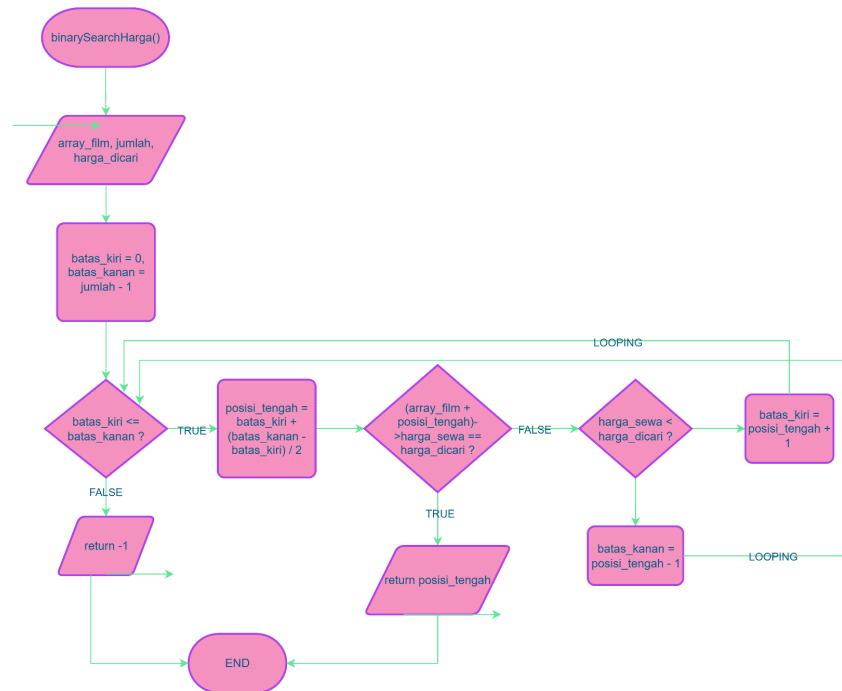


Gambar 1.12 Sorting Harga



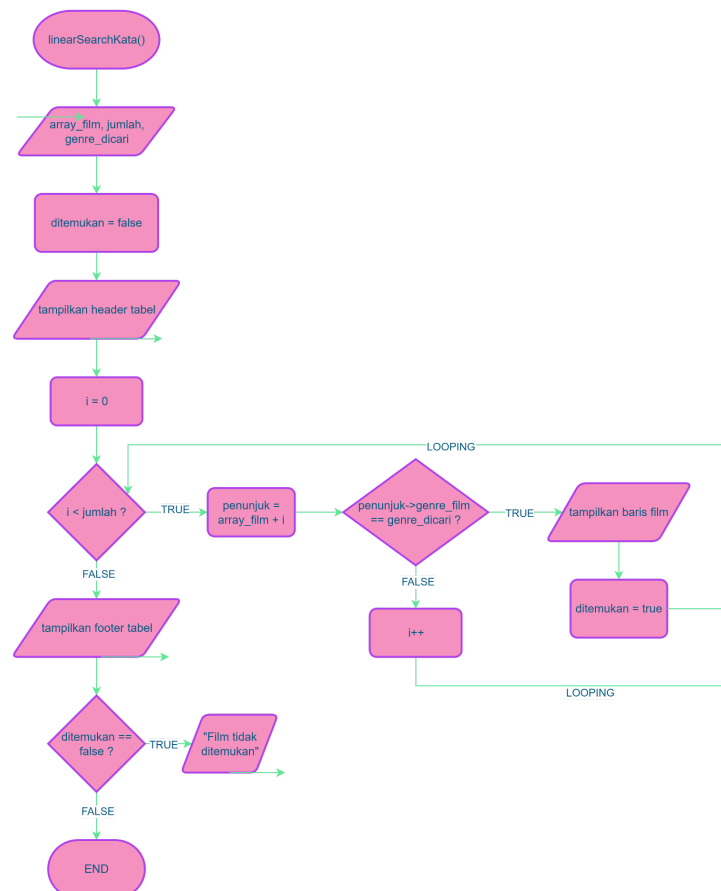
Gambar 1.13 Sorting Tahun

BINARY SEARCH



Gambar 1.14 Searching Harga

LINEAR SEARCH



Gambar 1.15 Searching Genre

1.2 Penjelasan Flowchart

Program CinemaRent diawali dengan inisialisasi data film dan pengguna, kemudian menampilkan menu utama dengan pilihan login, register, dan keluar. Proses login ditangani oleh fungsi `prosesLogin()` dengan batas tiga kali percobaan menggunakan pass by reference, dan jika berhasil pengguna diarahkan ke menu sesuai role yaitu Menu User atau Menu Admin. Menu User menyediakan empat fitur yaitu lihat film, sewa film, kembalikan film, dan riwayat sewa, sedangkan Menu Admin memiliki fitur tambahan berupa tambah, edit, hapus film, dan semua riwayat. Pada fitur Lihat Film, pengguna terlebih dahulu memilih metode sorting antara Bubble Sort untuk judul A-Z, Selection Sort untuk harga tertinggi, atau Insertion Sort untuk tahun terlama, setelah tabel ditampilkan pengguna dapat memilih aksi lanjutan yaitu melihat detail film by kode, mencari film by harga menggunakan Binary Search, atau mencari film by genre menggunakan Linear Search. Fungsi `binarySearchHarga()` menerima parameter pointer `Film*` dan bekerja dengan membagi array yang sudah terurut menjadi dua bagian di setiap iterasi untuk menemukan film berdasarkan harga, sedangkan fungsi `linearSearchKata()` menerima parameter pointer `Film*` dan menelusuri seluruh array satu per satu untuk menampilkan semua film yang genrenya sesuai dengan input pengguna. Fitur sewa menggunakan `inputTanggal()` dengan pass by reference untuk mencatat tanggal sewa dan menghitung tanggal wajib kembali, lalu data disimpan melalui prosedur `simpanDataSewa()` menggunakan pointer `DataSewa*`. Fitur kembalikan film menghitung selisih hari keterlambatan dan denda menggunakan fungsi rekursif `hitungDendaRekursif()`, serta menerapkan function overloading pada `hitungDenda()`. Pada Posttest 8, seluruh titik rawan input pada setiap fungsi menu ditambahkan mekanisme exception handling menggunakan blok try, throw, dan catch, dengan jenis exception yang digunakan meliputi `runtime_error`, `invalid_argument`, `out_of_range`, dan `length_error` sesuai konteks kesalahannya. Validasi input dikelola melalui user-defined library `validasi.h` yang berisi fungsi `validasiTidakKosong()`, `validasiRentangInt()`, `validasiMinInt()`, dan `validasiInputCin()`, sehingga program tidak crash ketika menerima input yang tidak valid dari pengguna. Dalam flowchart digunakan simbol terminator untuk START dan END, jajar genjang untuk input dan output, belah ketupat untuk percabangan, kotak biasa untuk proses, dan predefined process untuk setiap pemanggilan fungsi atau prosedur.

2. Deskripsi Singkat Program

Program CinemaRent merupakan aplikasi berbasis console menggunakan bahasa C++ untuk mengelola penyewaan film dengan dua jenis pengguna yaitu User dan Admin. User dapat melihat daftar film, menyewa, mengembalikan, dan melihat riwayat sewa, sedangkan Admin memiliki fitur tambahan untuk mengelola data film dan melihat seluruh riwayat transaksi. Pada Posttest 8, program dikembangkan dengan menerapkan error handling menggunakan exception handling melalui blok try, throw, dan catch pada seluruh fungsi menu yang berpotensi menerima input tidak valid, seperti login, register, sewa film, kembalikan film, tambah film, edit film, dan hapus film. Jenis exception yang digunakan meliputi `runtime_error` untuk kesalahan logika saat runtime seperti film tidak ditemukan atau nama sudah digunakan, `invalid_argument` untuk argumen yang tidak valid seperti input kosong atau nilai di bawah batas minimum, `out_of_range` untuk nilai di luar rentang yang diizinkan seperti hari dan bulan tanggal, serta `length_error` untuk kondisi kapasitas array yang sudah penuh. Selain itu, program dilengkapi dengan user-defined library pada file `validasi.h` yang di-include menggunakan tanda kutip `#include "validasi.h"`, berisi empat fungsi validasi yaitu `validasiTidakKosong()`, `validasiRentangInt()`, `validasiMinInt()`, dan `validasiInputCin()` yang dapat digunakan kembali di seluruh bagian program, sehingga memenuhi poin plus penggunaan user-defined library yang disyaratkan.

3. Source Code

3.1 Deklarasi Struct & Variabel Global

```
#include <iostream>
#include <string>
#include <iomanip>
#include <stdexcept>
#include "validasi.h"
using namespace std;

struct Studio {
    string nama_studio;
    string negara_studio;
};

struct Film {
    string kode_film;
    string judul_film;
    string genre_film;
    int    tahun_rilis;
    int    harga_sewa;
    Studio studio;
    bool    tersedia;
};

struct Pengguna {
    string nama_pengguna;
    string password_pengguna;
    string role_pengguna;
};

struct Tanggal {
    int hari;
    int bulan;
    int tahun;
};

struct DataSewa {
    int    id_sewa;
    string password_penyewa;
    string nama_penyewa;
    string kode_film_sewa;
    string judul_film_sewa;
    int    harga_per_hari;
    int    durasi_hari;
    Tanggal tanggal_sewa;
    Tanggal tanggal_kembali;
```

```

    int    total_bayar;
    int    denda;
    bool    sudah_kembali;
};

#define MAKS_FILM      50
#define MAKS_USER      20
#define MAKS_SEWA      100
#define DENDA_PER_HARI 5000

Film    daftar_film[MAKS_FILM];
int      jumlah_film = 0;

Pengguna daftar_user[MAKS_USER];
int      jumlah_user = 0;

DataSewa daftar_sewa[MAKS_SEWA];
int      jumlah_sewa  = 0;
int      id_berikutnya = 1;

string nama_aktif      = "";
string password_aktif  = "";
string role_aktif      = "";

```

3.2 Deklarasi Fungsi & Prosedur

```

void inisialisasiData();
void menuLihatFilm();
void menuSewaFilm();
void menuKembalikanFilm();
void menuRiwayatSewa(string nama_pengguna, string password_pengguna);
void menuTambahFilm();
void menuEditFilm();
void menuHapusFilm();
void menuSemuaRiwayat();
void prosesRegister(string role_default);
void inputTanggal(int &hari, int &bulan, int &tahun);

bool prosesLogin(string &nama_hasil, string &password_hasil, string
&role_hasil);
int  hitungTotalBayar(int harga_per_hari, int durasi_hari);
int  hitungDendaRekursif(int selisih_hari);
int  hitungDenda(int selisih_hari);
int  hitungDenda(int harga_per_hari, int selisih_hari);
int  cariFilm(string kode_film);
int  cariSewa(int id_sewa, string password penyewa);
int  hitungSelisihHari(Tanggal tanggal_wajib, Tanggal tanggal_aktual);

```

```

void tampilDetailFilm(Film *penunjuk_film);
void simpanDataSewa(DataSewa *penunjuk_sewa, int id_baru, string kode_film,
                  string judul_film, int harga_hari, int durasi,
                  Tanggal tgl_sewa, Tanggal tgl_kembali, int total);

void sortingJudul();
void sortingHarga();
void sortingTahun();

int binarySearchHarga(Film *array_film, int jumlah, int harga_dicari);
void linearSearchKata(Film *array_film, int jumlah, string genre_dicari);

```

3.3 User-Defined Library (validasi.h)

```

#ifndef VALIDASI_H
#define VALIDASI_H

#include <iostream>
#include <string>
#include <stdexcept>
using namespace std;

void validasiTidakKosong(string teks, string nama_field) {
    if (teks.empty()) {
        throw invalid_argument(nama_field + " tidak boleh kosong!");
    }
}

void validasiRentangInt(int nilai, int batas_bawah, int batas_atas,
string nama_field) {
    if (nilai < batas_bawah || nilai > batas_atas) {
        throw out_of_range(nama_field + " harus antara " +
to_string(batas_bawah) + " dan " + to_string(batas_atas) + "!");
    }
}

void validasiMinInt(int nilai, int batas_bawah, string nama_field) {
    if (nilai < batas_bawah) {
        throw invalid_argument(nama_field + " minimal " +
to_string(batas_bawah) + "!");
    }
}

void validasiInputCin() {

```



```

        if (cin.fail()) {
            cin.clear();
            cin.ignore(1000, '\n');
            throw runtime_error("Input harus berupa angka!");
        }
    }
}

#endif

```

3.4 Fitur Inisialisasi Data

```

void inisialisasiData() {
    daftar_user[0] = {"Admin", "000", "Admin"};
    daftar_user[1] = {"Nurul", "2509106033", "User"};
    jumlah_user = 2;

    daftar_film[0] = {"F001", "The Nun II", "Horror", 2023,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[1] = {"F002", "Hereditary", "Horror", 2018,
12000, {"A24", "Amerika"}, true};
    daftar_film[2] = {"F003", "It (2017)", "Horror", 2017,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[3] = {"F004", "Talk to Me", "Horror", 2023,
12000, {"A24", "Amerika"}, true};
    daftar_film[4] = {"F005", "Your Name", "Romance", 2016,
10000, {"Toho", "Jepang"}, true};
    daftar_film[5] = {"F006", "To All The Boys", "Romance", 2018,
10000, {"Netflix", "Amerika"}, true};
    daftar_film[6] = {"F007", "Five Feet Apart", "Romance", 2019,
10000, {"Warner Bros", "Amerika"}, true};
    daftar_film[7] = {"F008", "A Silent Voice", "Romance", 2016,
10000, {"Toho", "Jepang"}, true};
    daftar_film[8] = {"F009", "The Dark Knight", "Action", 2008,
13000, {"Warner Bros", "Amerika"}, true};
    daftar_film[9] = {"F010", "John Wick", "Action", 2014,
13000, {"Lionsgate", "Amerika"}, true};
    daftar_film[10] = {"F011", "Mad Max: Fury Road", "Action", 2015,
13000, {"Warner Bros", "Amerika"}, true};
    daftar_film[11] = {"F012", "Demon Slayer: Mugen", "Action", 2020,
13000, {"Toho", "Jepang"}, true};
    daftar_film[12] = {"F013", "Saltburn", "Thriller", 2023,
12000, {"A24", "Amerika"}, true};
    daftar_film[13] = {"F014", "Gone Girl", "Thriller", 2014,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[14] = {"F015", "Bird Box", "Thriller", 2018,
12000, {"Netflix", "Amerika"}, true};
}

```

```

    daftar_film[15] = {"F016", "Midsommar", "Thriller", 2019,
12000, {"A24", "Amerika"}, true};
    daftar_film[16] = {"F017", "Super Mario Bros Movie", "Comedy", 2023,
10000, {"Nintendo", "Amerika"}, true};
    daftar_film[17] = {"F018", "Puss in Boots 2", "Comedy", 2022,
10000, {"DreamWorks", "Amerika"}, true};
    daftar_film[18] = {"F019", "How To Train Your Dragon", "Animation", 2014,
10000, {"DreamWorks", "Amerika"}, true};
    daftar_film[19] = {"F020", "Spirited Away", "Animation", 2001,
10000, {"Toho", "Jepang"}, true};
    jumlah_film = 20;
}

```

3.5 Fitur Login

```

bool prosesLogin(string &nama_hasil, string &password_hasil, string &role_hasil)
{
    int sisa_percobaan = 3;
    bool login_berhasil = false;
    cin.ignore();
    while (sisa_percobaan > 0 && !login_berhasil) {
        try {
            string input_nama, input_password;
            cout << " Nama : "; getline(cin, input_nama);
            cout << " Password : "; getline(cin, input_password);
            validasiTidakKosong(input_nama, "Nama");
            validasiTidakKosong(input_password, "Password");
            bool cocok = false;
            for (int i = 0; i < jumlah_user; i++) {
                if (daftar_user[i].nama_pengguna == input_nama &&
                    daftar_user[i].password_pengguna == input_password) {
                    nama_hasil = daftar_user[i].nama_pengguna;
                    password_hasil = daftar_user[i].password_pengguna;
                    role_hasil = daftar_user[i].role_pengguna;
                    cocok = true;
                    break;
                }
            }
            if (!cocok) throw runtime_error("Nama atau password salah.");
            login_berhasil = true;
            cout << " [v] Selamat datang, " << nama_hasil << "!" << endl;
            cout << " [Tekan Enter...]" << cin.get();
        } catch (exception &e) {
            sisa_percobaan--;
            cout << " [!] " << e.what();
            if (sisa_percobaan > 0)
                cout << " Sisa percobaan: " << sisa_percobaan << endl;
            else

```

```

        cout << endl << " [!!!] Gagal 3 kali. Kembali ke menu utama."
<< endl;
    }
}
return login_berhasil;
}

```

3.6 Fitur Register

```

void prosesRegister(string role_default) {
    try {
        if (jumlah_user >= MAKS_USER) throw length_error("Kapasitas pengguna
penuh.");
        cin.ignore();
        string nama_baru, password_baru;
        cout << " Nama      : "; getline(cin, nama_baru);
        validasiTidakKosong(nama_baru, "Nama");
        for (int i = 0; i < jumlah_user; i++) {
            if (daftar_user[i].nama_pengguna == nama_baru) throw
runtime_error("Nama sudah digunakan.");
        }
        cout << " Password : "; getline(cin, password_baru);
        validasiTidakKosong(password_baru, "Password");
        daftar_user[jumlah_user] = {nama_baru, password_baru, role_default};
        jumlah_user++;
        cout << " [v] Akun berhasil dibuat!" << endl;
    } catch (exception &e) {
        cout << " [!] " << e.what() << endl;
    }
    cout << " [Tekan Enter...]"; cin.get();
}

```

3.7 Fitur Lihat Film & Tampil Detail Film

```

void inputTanggal(int &hari, int &bulan, int &tahun) {
    bool input_valid = false;
    while (!input_valid) {
        try {
            cout << " Hari (1-31) : ";
            cin >> hari;
            validasiInputCin();
            validasiRentangInt(hari, 1, 31, "Hari");
            input_valid = true;
        } catch (exception &e) {
            cout << " [!] " << e.what() << endl;
        }
    }
}

```

```

    }
    input_valid = false;
    while (!input_valid) {
        try {
            cout << "    Bulan (1-12) : ";
            cin >> bulan;
            validasiInputCin();
            validasiRentangInt(bulan, 1, 12, "Bulan");
            input_valid = true;
        } catch (exception &e) {
            cout << "    [!] " << e.what() << endl;
        }
    }
    input_valid = false;
    while (!input_valid) {
        try {
            cout << "    Tahun          : ";
            cin >> tahun;
            validasiInputCin();
            validasiMinInt(tahun, 2000, "Tahun");
            input_valid = true;
        } catch (exception &e) {
            cout << "    [!] " << e.what() << endl;
        }
    }
}

void menuLihatFilm() {
    int pilihan_sort = 0;
    bool pilihan_valid = false;
    while (!pilihan_valid) {
        try {
            cout << "    Urutkan berdasarkan:" << endl;
            cout << "    1. Judul A-Z          (Bubble Sort)" << endl;
            cout << "    2. Harga Tertinggi (Selection Sort)" << endl;
            cout << "    3. Tahun Terlama   (Insertion Sort)" << endl;
            cout << "    Pilihan : ";
            cin >> pilihan_sort;
            validasiInputCin();
            validasiRentangInt(pilihan_sort, 1, 3, "Pilihan");
            pilihan_valid = true;
        } catch (exception &e) {
            cout << "    [!] " << e.what() << endl;
        }
    }

    if      (pilihan_sort == 1) sortingJudul();
    else if (pilihan_sort == 2) sortingHarga();
    else if (pilihan_sort == 3) sortingTahun();

    cout <<

```

```

"+-----+-----+-----+-----+-----+-----+-----+-----+-----+
-+" << endl;
    cout << " | Kode | Judul | Genre | Tahun |
Harga/Hr | Status |" << endl;
    cout <<
"+-----+-----+-----+-----+-----+-----+-----+-----+-----+
-+" << endl;
    for (int i = 0; i < jumlah_film; i++) {
        string status_film = daftar_film[i].tersedia ? "Tersedia" : "Disewa ";
        string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << " | " << left << setw(5) << daftar_film[i].kode_film
            << " | " << setw(29) << daftar_film[i].judul_film
            << " | " << setw(10) << daftar_film[i].genre_film
            << " | " << setw(5) << daftar_film[i].tahun_rilis
            << " | " << setw(8) << harga_str
            << " | " << status_film << "|" << endl;
    }
    cout <<
"+-----+-----+-----+-----+-----+-----+-----+-----+-----+
-+" << endl;

    cout << endl;
    cout << " Pilih aksi:" << endl;
    cout << " 1. Lihat detail film (by kode)" << endl;
    cout << " 2. Cari film by harga [Binary Search]" << endl;
    cout << " 3. Cari film by genre [Linear Search]" << endl;
    cout << " 0. Kembali" << endl;

    int pilihan_aksi = 0;
    pilihan_valid = false;
    while (!pilihan_valid) {
        try {
            cout << " Pilihan : ";
            cin >> pilihan_aksi;
            validasiInputCin();
            validasiRentangInt(pilihan_aksi, 0, 3, "Pilihan");
            pilihan_valid = true;
        } catch (exception &e) {
            cout << " [!] " << e.what() << endl;
        }
    }

    if (pilihan_aksi == 1) {
        try {
            string kode_detail;
            cout << " Kode film : ";
            cin.ignore(); getline(cin, kode_detail);
            validasiTidakKosong(kode_detail, "Kode film");
            int indeks_detail = cariFilm(kode_detail);
            if (indeks_detail == -1) throw runtime_error("Film tidak
ditemukan.");

```

```

        Film *penunjuk_film = &daftar_film[indeks_detail];
        tampilDetailFilm(penunjuk_film);
    } catch (exception &e) {
        cout << "  [!] " << e.what() << endl;
    }
    cout << "  [Tekan Enter...]" << cin.get();

} else if (pilihan_aksi == 2) {
    Film salinan_film[MAKS_FILM];
    for (int i = 0; i < jumlah_film; i++) salinan_film[i] = daftar_film[i];
    for (int i = 1; i < jumlah_film; i++) {
        Film film_kunci = salinan_film[i];
        int j = i - 1;
        while (j >= 0 && salinan_film[j].harga_sewa > film_kunci.harga_sewa)
        {
            salinan_film[j + 1] = salinan_film[j];
            j--;
        }
        salinan_film[j + 1] = film_kunci;
    }
    try {
        int harga_dicari = 0;
        cout << "  Masukkan harga yang ingin dicari (Rp) : ";
        cin >> harga_dicari;
        validasiInputCin();
        validasiMinInt(harga_dicari, 1, "Harga");
        int indeks_hasil = binarySearchHarga(salinan_film, jumlah_film,
harga_dicari);
        if (indeks_hasil == -1) throw runtime_error("Film dengan harga Rp" +
to_string(harga_dicari) + " tidak ditemukan.");
        cout << "  [v] Film dengan harga Rp" << harga_dicari << "
ditemukan!" << endl;
        tampilDetailFilm(salinan_film + indeks_hasil);
        bool ada_lain = false;
        int kiri = indeks_hasil - 1;
        int kanan = indeks_hasil + 1;
        while (kiri >= 0 && (salinan_film + kiri)->harga_sewa ==
harga_dicari) {
            if (!ada_lain) { cout << "  Film lain dengan harga sama:" <<
endl; ada_lain = true; }
            tampilDetailFilm(salinan_film + kiri);
            kiri--;
        }
        while (kanan < jumlah_film && (salinan_film + kanan)->harga_sewa ==
harga_dicari) {
            if (!ada_lain) { cout << "  Film lain dengan harga sama:" <<
endl; ada_lain = true; }
            tampilDetailFilm(salinan_film + kanan);
            kanan++;
        }
    } catch (exception &e) {

```

```

        cout << "  [!] " << e.what() << endl;
    }
    cout << "  [Tekan Enter...]" << cin.ignore(); cin.get();

    } else if (pilihan_aksi == 3) {
        try {
            string genre_dicari;
            cout << "  Genre yang tersedia: Horror, Romance, Action, Thriller,
Comedy, Animation" << endl;
            cout << "  Masukkan genre yang dicari : ";
            cin.ignore(); getline(cin, genre_dicari);
            validasiTidakKosong(genre_dicari, "Genre");
            cout << "  Hasil pencarian genre \" " << genre_dicari << "\" : " <<
endl;

            linearSearchKata(daftar_film, jumlah_film, genre_dicari);
        } catch (exception &e) {
            cout << "  [!] " << e.what() << endl;
        }
        cout << "  [Tekan Enter...]" << cin.get();
    }
}
}

```

3.8 Fungsi Sorting

```

void sortingJudul() {
    Film film_tukar;
    bool sudah_urut;
    for (int i = 0; i < jumlah_film - 1; i++) {
        sudah_urut = false;
        for (int j = 0; j < jumlah_film - i - 1; j++) {
            if (daftar_film[j].judul_film > daftar_film[j +
1].judul_film) {
                film_tukar          = daftar_film[j];
                daftar_film[j]      = daftar_film[j + 1];
                daftar_film[j + 1] = film_tukar;
                sudah_urut          = true;
            }
        }
        if (!sudah_urut) break;
    }
}

void sortingHarga() {
    int indeks_terbesar;
    Film film_tukar;
    for (int i = 0; i < jumlah_film - 1; i++) {
        indeks_terbesar = i;
    }
}

```

```

        for (int j = i + 1; j < jumlah_film; j++) {
            if (daftar_film[j].harga_sewa >
daftar_film[indeks_terbesar].harga_sewa) {
                indeks_terbesar = j;
            }
        }
        if (indeks_terbesar != i) {
            film_tukar = daftar_film[i];
            daftar_film[i] = daftar_film[indeks_terbesar];
            daftar_film[indeks_terbesar] = film_tukar;
        }
    }
}

void sortingTahun() {
    Film film_kunci;
    int j;
    for (int i = 1; i < jumlah_film; i++) {
        film_kunci = daftar_film[i];
        j = i - 1;
        while (j >= 0 && daftar_film[j].tahun_rilis >
film_kunci.tahun_rilis) {
            daftar_film[j + 1] = daftar_film[j];
            j = j - 1;
        }
        daftar_film[j + 1] = film_kunci;
    }
}

```

3.9 Fungsi Searching

```

int binarySearchHarga(Film *array_film, int jumlah, int
harga_dicari) {
    int batas_kiri = 0;
    int batas_kanan = jumlah - 1;

    while (batas_kiri <= batas_kanan) {
        int posisi_tengah = batas_kiri + (batas_kanan -
batas_kiri) / 2;

        if ((array_film + posisi_tengah)->harga_sewa ==
harga_dicari) {
            return posisi_tengah;
        } else if ((array_film + posisi_tengah)->harga_sewa <

```



```

harga_dicari) {
    batas_kiri = posisi_tengah + 1;
} else {
    batas_kanan = posisi_tengah - 1;
}
}

return -1;
}

void linearSearchKata(Film *array_film, int jumlah, string
genre_dicari) {
    bool ditemukan = false;

    cout << "
+-----+-----+-----+-----+-----+-----+
----+-----+" << endl;
    cout << " | Kode | Judul | Genre
| Tahun | Harga/Hr | Status |" << endl;
    cout << "
+-----+-----+-----+-----+-----+-----+
----+-----+" << endl;

    for (int i = 0; i < jumlah; i++) {
        Film *penunjuk = array_film + i;

        if (penunjuk->genre_film == genre_dicari) {
            string status_film = penunjuk->tersedia ? "Tersedia" :
"Disewa ";
            string harga_str = "Rp" +
to_string(penunjuk->harga_sewa);
            cout << " | " << left << setw(5) <<
penunjuk->kode_film
                << " | " << setw(29) <<
penunjuk->judul_film
                << " | " << setw(10) <<
penunjuk->genre_film
                << " | " << setw(5) <<
penunjuk->tahun_rilis
                << " | " << setw(8) << harga_str
                << " | " << status_film << " | " << endl;
            ditemukan = true;

```

```

    }
}

cout << "
+-----+-----+-----+-----+-----+-----+
----+-----+" << endl;

if (!ditemukan) {
    cout << "  [] Film dengan genre \">

```

3.10 Fitur Sewa Film

```

void menuSewaFilm() {
    cout << "  === SEWA FILM ===" << endl;
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;
    cout << "| Kode   | Judul                                     | Genre   | Harga/Hr |" <<
endl;
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;

    bool ada_tersedia = false;
    for (int i = 0; i < jumlah_film; i++) {
        if (!daftar_film[i].tersedia) continue;
        string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << "| " << left << setw(5) << daftar_film[i].kode_film
            << "| "
            << setw(29) << daftar_film[i].judul_film
            << "| "
            << setw(10) << daftar_film[i].genre_film
            << "| "
            << setw(8) << harga_str << "|" << endl;
        ada_tersedia = true;
    }
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;

    if (!ada_tersedia) {
        cout << "  Semua film sedang disewa." << endl;
        cin.ignore(); cin.get(); return;
    }

    try {
        string kode_dipilih;

```

```

        cout << " Kode film (Enter -> Kembali) : ";
        cin.ignore(); getline(cin, kode_dipilih);
        if (kode_dipilih == "") return;
        validasiTidakKosong(kode_dipilih, "Kode film");

        int indeks_film = cariFilm(kode_dipilih);
        if (indeks_film == -1) throw runtime_error("Film tidak ditemukan.");
        if (!daftar_film[indeks_film].tersedia) throw runtime_error("Film sedang
disewa oleh pengguna lain.");
        if (jumlah_sewa >= MAKS_SEWA) throw length_error("Kapasitas data sewa
sudah penuh.");

        Tanggal tanggal_sewa;
        cout << " Tanggal Sewa:" << endl;
        inputTanggal(tanggal_sewa.hari, tanggal_sewa.bulan, tanggal_sewa.tahun);

        int durasi_hari = 0;
        bool durasi_valid = false;
        while (!durasi_valid) {
            try {
                cout << " Durasi (hari, min 1) : ";
                cin >> durasi_hari;
                validasiInputCin();
                validasiMinInt(durasi_hari, 1, "Durasi");
                durasi_valid = true;
            } catch (exception &e) {
                cout << " [!] " << e.what() << endl;
            }
        }

        Tanggal tanggal_kembali = tanggal_sewa;
        int hari_per_bulan[13] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31,
30, 31};
        tanggal_kembali.hari += durasi_hari;
        while (tanggal_kembali.hari > hari_per_bulan[tanggal_kembali.bulan]) {
            tanggal_kembali.hari -= hari_per_bulan[tanggal_kembali.bulan];
            tanggal_kembali.bulan++;
            if (tanggal_kembali.bulan > 12) { tanggal_kembali.bulan = 1;
tanggal_kembali.tahun++; }
        }

        int total_bayar_sewa =
hitungTotalBayar(daftar_film[indeks_film].harga_sewa, durasi_hari);

        cout << " Film : " << daftar_film[indeks_film].judul_film <<
endl;
        cout << " Tgl Sewa : ";
        if (tanggal_sewa.hari < 10) cout << "0"; cout << tanggal_sewa.hari <<
"/";
        if (tanggal_sewa.bulan < 10) cout << "0"; cout << tanggal_sewa.bulan <<
"/" << tanggal_sewa.tahun << endl;

```

```

        cout << "  Wajib Kembali : ";
        if (tanggal_kembali.hari < 10) cout << "0"; cout <<
tanggal_kembali.hari << "/";
        if (tanggal_kembali.bulan < 10) cout << "0"; cout <<
tanggal_kembali.bulan << "/" << tanggal_kembali.tahun << endl;
        cout << "  Total Bayar   : Rp" << total_bayar_sewa << endl;
        cout << "  Denda       : Rp" << DENDA_PER_HARI << "/hari jika
terlambat" << endl;
        cout << "  Konfirmasi? (y/n) : ";
        char konfirmasi_sewa; cin >> konfirmasi_sewa;
        if (konfirmasi_sewa != 'y' && konfirmasi_sewa != 'Y') {
            cout << "  Dibatalkan." << endl; cin.ignore(); cin.get(); return;
        }

        daftar_film[indeks_film].tersedia = false;
        DataSewa *penunjuk_sewa = &daftar_sewa[jumlah_sewa];
        simpanDataSewa(penunjuk_sewa, id_berikutnya++, kode_dipilih,
                        daftar_film[indeks_film].judul_film,
                        daftar_film[indeks_film].harga_sewa,
                        durasi_hari, tanggal_sewa, tanggal_kembali,
total_bayar_sewa);
        jumlah_sewa++;
        cout << "  [v] Berhasil menyewa \"" <<
daftar_film[indeks_film].judul_film << "\"!" << endl;

    } catch (exception &e) {
        cout << "  [!] " << e.what() << endl;
    }
    cout << "  [Tekan Enter...]" << endl; cin.ignore(); cin.get();
}

```

3.11 Fitur Kembalikan Film

```

void menuKembalikanFilm() {
    cout << "  === KEMBALIKAN FILM ===" << endl;
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;
    cout << "| ID | Judul                               | Tgl Sewa   | Wajib Kemb |" <<
endl;
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;

    bool ada_sewa_aktif = false;
    for (int i = 0; i < jumlah_sewa; i++) {
        if (daftar_sewa[i].password_penyewa != password_aktif ||
daftar_sewa[i].sudah_kembali) continue;
        string tgl_sewa_str = "";
        if (daftar_sewa[i].tanggal_sewa.hari < 10) tgl_sewa_str += "0";
    }
}

```

```

        tgl_sewa_str += to_string(daftar_sewa[i].tanggal_sewa.hari) + "/";
        if (daftar_sewa[i].tanggal_sewa.bulan < 10) tgl_sewa_str += "0";
        tgl_sewa_str += to_string(daftar_sewa[i].tanggal_sewa.bulan) + "/" +
to_string(daftar_sewa[i].tanggal_sewa.tahun);
        string tgl_kemb_str = "";
        if (daftar_sewa[i].tanggal_kembali.hari < 10) tgl_kemb_str += "0";
        tgl_kemb_str += to_string(daftar_sewa[i].tanggal_kembali.hari) + "/";
        if (daftar_sewa[i].tanggal_kembali.bulan < 10) tgl_kemb_str += "0";
        tgl_kemb_str += to_string(daftar_sewa[i].tanggal_kembali.bulan) + "/" +
to_string(daftar_sewa[i].tanggal_kembali.tahun);
        cout << "| " << left << setw(2) << daftar_sewa[i].id_sewa
        << " | " << setw(29) << daftar_sewa[i].judul_film_sewa
        << "| " << setw(11) << tgl_sewa_str
        << "| " << setw(11) << tgl_kemb_str << "|" << endl;
        ada_sewa_aktif = true;
    }
    cout << "+---+-----+-----+-----+-----+" <<
endl;

    if (!ada_sewa_aktif) {
        cout << " Tidak ada sewa aktif." << endl; cin.ignore(); cin.get();
return;
    }

    try {
        int id_dikembalikan = 0;
        bool id_valid = false;
        while (!id_valid) {
            try {
                cout << " ID sewa : ";
                cin >> id_dikembalikan;
                validasiInputCin();
                validasiMinInt(id_dikembalikan, 1, "ID sewa");
                id_valid = true;
            } catch (exception &e) {
                cout << " [!] " << e.what() << endl;
            }
        }

        int indeks_sewa = cariSewa(id_dikembalikan, password_aktif);
        if (indeks_sewa == -1) throw runtime_error("ID sewa tidak valid atau
bukan milik akun ini.");

        Tanggal tanggal_aktual_kembali;
        cout << " Tanggal Kembali:" << endl;
        inputTanggal(tanggal_aktual_kembali.hari, tanggal_aktual_kembali.bulan,
tanggal_aktual_kembali.tahun);

        int selisih_hari =
hitungSelisihHari(daftar_sewa[indeks_sewa].tanggal_kembali,
tanggal_aktual_kembali);

```

```

    int denda_terlambat = hitungDenda(selisih_hari);

    daftar_sewa[indeks_sewa].denda = denda_terlambat;
    daftar_sewa[indeks_sewa].sudah_kembali = true;

    int indeks_film_kembali =
    cariFilm(daftar_sewa[indeks_sewa].kode_film_sewa);
    if (indeks_film_kembali != -1) daftar_film[indeks_film_kembali].tersedia
    = true;

    cout << " [v] Film berhasil dikembalikan!" << endl;
    if (selisih_hari > 0) {
        cout << " Terlambat " << selisih_hari << " hari" << endl;
        cout << " Denda : Rp" << denda_terlambat << endl;
    } else {
        cout << " Tepat waktu, tidak ada denda." << endl;
    }
    cout << " TOTAL BAYAR: Rp" << daftar_sewa[indeks_sewa].total_bayar +
    denda_terlambat << endl;

    } catch (exception &e) {
        cout << " [!] " << e.what() << endl;
    }
    cout << " [Tekan Enter...]"; cin.ignore(); cin.get();
}

```

3.12 Fitur Riwayat Sewa

```

void menuRiwayatSewa(string nama_pengguna, string password_pengguna) {
    cout << " === RIWAYAT SEWA - " << nama_pengguna << " ===" << endl;
    cout << "+---+-----+-----+-----+-----+"
<< endl;
    cout << "| ID | Judul | Total | Status |"
<< endl;
    cout << "+---+-----+-----+-----+-----+"
<< endl;

    bool ada_riwayat = false;
    for (int i = 0; i < jumlah_sewa; i++) {
        if (daftar_sewa[i].password_penyewa != password_pengguna)
        continue;
        string status_sewa = daftar_sewa[i].sudah_kembali ? "Selesai "
: "Aktif ";
        int total_tampil = daftar_sewa[i].total_bayar +
        daftar_sewa[i].denda;
    }
}

```

```

        string total_str      = "Rp" + to_string(total_tampil);
        cout << "|" << left << setw(2) << daftar_sewa[i].id_sewa
              << " | "      << setw(29) <<
daftar_sewa[i].judul_film_sewa
              << "|" <<      << setw(8) << total_str
              << " | "      << status_sewa << "|" << endl;
        ada_riwayat = true;
    }
    cout << "+-----+-----+-----+-----+-----+"
<< endl;
    if (!ada_riwayat) cout << "  Belum ada riwayat." << endl;
    cout << "  [Tekan Enter...]" ; cin.ignore(); cin.get();
}

```

3.13 Fitur Tambah Film (Admin)

```

void menuTambahFilm() {
    cout << "  === TAMBAH FILM ===" << endl;
    try {
        if (jumlah_film >= MAKS_FILM) throw length_error("Kapasitas film sudah
penuh.");

        cout <<
"+-----+-----+-----+-----+-----+" <<
endl;
        cout << " | Kode   | Judul                               | Genre   | Tahun |
Harga/Hr |" << endl;
        cout <<
"+-----+-----+-----+-----+-----+" <<
endl;
        for (int i = 0; i < jumlah_film; i++) {
            string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
            cout << " | " << left << setw(5) << daftar_film[i].kode_film
                  << " | "      << setw(29) << daftar_film[i].judul_film
                  << " | "      << setw(10) << daftar_film[i].genre_film
                  << " | "      << setw(5) << daftar_film[i].tahun_rilis
                  << " | "      << setw(8) << harga_str << "|" << endl;
        }
        cout <<
"+-----+-----+-----+-----+-----+" <<
endl;

        cin.ignore();

        string kode_baru;
        cout << "  Kode   : "; getline(cin, kode_baru);
    }
}

```

```

    if (kode_baru == "") return;
    validasiTidakKosong(kode_baru, "Kode film");

    for (int i = 0; i < jumlah_film; i++) {
        if (daftar_film[i].kode_film == kode_baru) throw runtime_error("Kode
film sudah ada.");
    }

    string judul_baru, genre_baru, studio_baru, negara_baru;
    int tahun_baru = 0, harga_baru = 0;

    cout << "   Judul   : "; getline(cin, judul_baru);
    validasiTidakKosong(judul_baru, "Judul");

    cout << "   Genre   : "; getline(cin, genre_baru);
    validasiTidakKosong(genre_baru, "Genre");

    bool tahun_valid = false;
    while (!tahun_valid) {
        try {
            cout << "   Tahun   : "; cin >> tahun_baru;
            validasiInputCin();
            validasiMinInt(tahun_baru, 1888, "Tahun");
            tahun_valid = true;
        } catch (exception &e) {
            cout << "   [!]   " << e.what() << endl;
        }
    }

    bool harga_valid = false;
    while (!harga_valid) {
        try {
            cout << "   Harga   : "; cin >> harga_baru;
            validasiInputCin();
            validasiMinInt(harga_baru, 1, "Harga");
            harga_valid = true;
        } catch (exception &e) {
            cout << "   [!]   " << e.what() << endl;
        }
    }

    cin.ignore();
    cout << "   Studio  : "; getline(cin, studio_baru);
    validasiTidakKosong(studio_baru, "Studio");

    cout << "   Negara  : "; getline(cin, negara_baru);
    validasiTidakKosong(negara_baru, "Negara");

    daftar_film[jumlah_film] = {kode_baru, judul_baru, genre_baru,
tahun_baru, harga_baru, {studio_baru, negara_baru}, true};
    jumlah_film++;
    cout << "   [v]   Film berhasil ditambahkan!" << endl;

```



```

    } catch (exception &e) {
        cout << "  [!] " << e.what() << endl;
    }
    cout << "  [Tekan Enter...]" ; cin.get();
}

```

3.14 Fitur Edit Film (Admin)

```

void menuEditFilm() {
    cout << "  === EDIT FILM ===" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+" <<
    endl;
    cout << " | Kode   | Judul                               | Genre   | Tahun   |
    Harga/Hr |" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+" <<
    endl;
    for (int i = 0; i < jumlah_film; i++) {
        string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << " | " << left << setw(5) << daftar_film[i].kode_film
        << " | " << setw(29) << daftar_film[i].judul_film
        << " | " << setw(10) << daftar_film[i].genre_film
        << " | " << setw(5) << daftar_film[i].tahun_rilis
        << " | " << setw(8) << harga_str << "|" << endl;
    }
    cout <<
    "+-----+-----+-----+-----+-----+" <<
    endl;

    try {
        cin.ignore();
        string kode_diedit;
        cout << "  Kode film (Enter -> Kembali) : ";
        getline(cin, kode_diedit);
        if (kode_diedit == "") return;
        validasiTidakKosong(kode_diedit, "Kode film");

        int indeks_edit = cariFilm(kode_diedit);
        if (indeks_edit == -1) throw runtime_error("Kode film tidak
        ditemukan.");

        cout << "  [1]Judul  [2]Genre  [3]Tahun  [4]Harga  [5]Studio  [6]Negara"

```

```

<< endl;

    int pilihan_edit = 0;
    bool edit_valid = false;
    while (!edit_valid) {
        try {
            cout << "    Pilihan    : "; cin >> pilihan_edit;
            validasiInputCin();
            validasiRentangInt(pilihan_edit, 1, 6, "Pilihan");
            edit_valid = true;
        } catch (exception &e) {
            cout << "    [!]    " << e.what() << endl;
        }
    }

    cin.ignore();
    if (pilihan_edit == 1) {
        string judul_baru;
        cout << "    Judul baru    : "; getline(cin, judul_baru);
        validasiTidakKosong(judul_baru, "Judul");
        daftar_film[indeks_edit].judul_film = judul_baru;
    } else if (pilihan_edit == 2) {
        string genre_baru;
        cout << "    Genre baru    : "; getline(cin, genre_baru);
        validasiTidakKosong(genre_baru, "Genre");
        daftar_film[indeks_edit].genre_film = genre_baru;
    } else if (pilihan_edit == 3) {
        int tahun_baru = 0;
        bool tahun_valid = false;
        while (!tahun_valid) {
            try {
                cout << "    Tahun baru    : "; cin >> tahun_baru;
                validasiInputCin();
                validasiMinInt(tahun_baru, 1888, "Tahun");
                tahun_valid = true;
            } catch (exception &e) {
                cout << "    [!]    " << e.what() << endl;
            }
        }
        daftar_film[indeks_edit].tahun_rilis = tahun_baru;
    } else if (pilihan_edit == 4) {
        int harga_baru = 0;
        bool harga_valid = false;
        while (!harga_valid) {
            try {
                cout << "    Harga baru    : "; cin >> harga_baru;
                validasiInputCin();
                validasiMinInt(harga_baru, 1, "Harga");
                harga_valid = true;
            } catch (exception &e) {
                cout << "    [!]    " << e.what() << endl;
            }
        }
    }
}

```

```

    }
}
daftar_film[indeks_edit].harga_sewa = harga_baru;
} else if (pilihan_edit == 5) {
    string studio_baru;
    cout << "   Studio baru : "; getline(cin, studio_baru);
    validasiTidakKosong(studio_baru, "Studio");
    daftar_film[indeks_edit].studio.nama_studio = studio_baru;
} else if (pilihan_edit == 6) {
    string negara_baru;
    cout << "   Negara baru : "; getline(cin, negara_baru);
    validasiTidakKosong(negara_baru, "Negara");
    daftar_film[indeks_edit].studio.negara_studio = negara_baru;
}
cout << "   [v] Film berhasil diedit!" << endl;

} catch (exception &e) {
    cout << "   [!] " << e.what() << endl;
}
cout << "   [Tekan Enter...]"; cin.get();
}

```

3.15 Fitur Hapus Film (Admin)

```

void menuHapusFilm() {
    cout << "   === HAPUS FILM ===" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+" <<
    endl;
    cout << " | Kode   | Judul                               | Genre   | Tahun | Status
    |" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+" <<
    endl;
    for (int i = 0; i < jumlah_film; i++) {
        string status_film = daftar_film[i].tersedia ? "Tersedia" : "Disewa  ";
        cout << " | " << left << setw(5) << daftar_film[i].kode_film
            << " | " << setw(29) << daftar_film[i].judul_film
            << " | " << setw(10) << daftar_film[i].genre_film
            << " | " << setw(5) << daftar_film[i].tahun_rilis
            << " | " << status_film << "|" << endl;
    }
    cout <<
    "+-----+-----+-----+-----+-----+" <<
    endl;

    try {

```



```

: hitungDenda(daftar_sewa[i].harga_per_hari, 0);
int total_tampil = daftar_sewa[i].total_bayar + denda_tampil;
string total_str = "Rp" + to_string(total_tampil);
cout << "| " << left << setw(2) << daftar_sewa[i].id_sewa
      << " | " << setw(10) << daftar_sewa[i].nama_penyewa
      << " | " << setw(29) << daftar_sewa[i].judul_film_sewa
      << " | " << setw(8) << total_str
      << " | " << status_sewa << "|" << endl;
}
cout <<
"+---+-----+-----+-----+-----+-----+" <<
endl;
cout << " [Tekan Enter...]"; cin.ignore(); cin.get();
}

```

3.17 Main Program

```

int main() {
    inisialisasiData();
    bool program_berjalan = true;
    do {
        cout << "===== " << endl;
        cout << "          SELAMAT DATANG DI CINEMARENT          " << endl;
        cout << "  1. Login " << endl;
        cout << "  2. Register " << endl;
        cout << "  0. Keluar " << endl;
        cout << "===== " << endl;

        int pilihan_awal = -1;
        try {
            cout << "  Pilihan : ";
            cin >> pilihan_awal;
            validasiInputCin();
            validasiRentangInt(pilihan_awal, 0, 2, "Pilihan");
        } catch (exception &e) {
            cout << "  [!] " << e.what() << endl;
            cout << "  [Tekan Enter...]"; cin.ignore(); cin.get();
            continue;
        }

        if (pilihan_awal == 2) {
            prosesRegister("User");
        } else if (pilihan_awal == 1) {
            bool login_ok = prosesLogin(nama_aktif, password_aktif, role_aktif);
            if (login_ok) {
                bool sedang_login = true;
                do {

```

```

        cout << "===== " << endl;

        cout << "  MENU  " << (role_aktif == "Admin" ? "ADMIN" :
"USER")

        << " | " << nama_aktif << endl;
        cout << "  1. Lihat Film" << endl;
        cout << "  2. Sewa Film" << endl;
        cout << "  3. Kembalikan" << endl;
        cout << "  4. Riwayat Sewa" << endl;
        if (role_aktif == "Admin") {
            cout << "-----" << endl;

            cout << "  5. Tambah Film" << endl;
            cout << "  6. Edit Film" << endl;
            cout << "  7. Hapus Film" << endl;
            cout << "  8. Semua Riwayat" << endl;
        }
        cout << "  0. Logout" << endl;
        cout << "-----" << endl;

        int pilihan_menu = -1;
        try {
            cout << "  Pilihan : ";
            cin >> pilihan_menu;
            validasiInputCin();
        } catch (exception &e) {
            cout << "  [!] " << e.what() << endl;
            cout << "  [Tekan Enter...]" << cin.ignore(); cin.get();
            continue;
        }

        switch (pilihan_menu) {
            case 1: menuLihatFilm();
            break;

            case 2: menuSewaFilm();
            break;

            case 3: menuKembalikanFilm();
            break;

            case 4: menuRiwayatSewa(nama_aktif, password_aktif);
            break;

            case 5:
                if (role_aktif == "Admin") menuTambahFilm();
                else { cout << "  [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                break;

            case 6:
                if (role_aktif == "Admin") menuEditFilm();
                else { cout << "  [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                break;

```

```

        case 7:
            if (role_aktif == "Admin") menuHapusFilm();
            else { cout << "  [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
            break;
        case 8:
            if (role_aktif == "Admin") menuSemuaRiwayat();
            else { cout << "  [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
            break;
        case 0:
            cout << "  Sampai jumpa, " << nama_aktif << "!" <<
endl;
            nama_aktif = ""; password_aktif = ""; role_aktif =
"";
            sedang_login = false;
            cout << "  [Tekan Enter...]" << cin.ignore();
cin.get();
            break;
        default:
            cout << "  [!] Pilihan tidak valid." << endl;
            cout << "  [Tekan Enter...]" << cin.ignore();
cin.get();
            break;
    }
} while (sedang_login);
}
} else if (pilihan_awal == 0) {
    cout << "  Terima kasih! Sampai jumpa!" << endl;
    program_berjalan = false;
}
} while (program_berjalan);
return 0;
}

```

4. Hasil Output

```
=====
                SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====
Pilihan : █
```

Gambar 4.1 Menu Login

```
=====
                SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====
Pilihan : 1
Nama      : Nana
Password  : 009
[!] Salah. Sisa percobaan: 2

Nama      : Nala
Password  : 008
[!] Salah. Sisa percobaan: 1

Nama      : Nurul
Password  : 2509106044
[!!!] Gagal 3 kali. Program berhenti.
```

Gambar 4.2 Login Gagal

```
=====
                SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====
Pilihan : 1
Nama      : Nurul
Password  : 2509106033
[v] Selamat datang, Nurul!
[Tekan Enter...] █
```

Gambar 4.3 Login Berhasil

```
=====
                SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====
Pilihan : 0
Terima kasih! Sampai jumpa!
PS C:\Users\LOQ\Documents\praktikum-apl\post-test\post-test-apl-2> █
```

Gambar 4.4 Keluar


```
=====
                        SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====

Pilihan : 2
Nama      : Nana
Password  : 009
[v] Akun berhasil dibuat!
[Tekan Enter...]
```

Gambar 4.5 Register

```
=====
MENU USER | Nurul
1. Lihat Film
2. Sewa Film
3. Kembalikan
4. Riwayat Sewa
0. Logout
=====

Pilihan : █
```

Gambar 4.6 Menu User

```
=====
Pilihan : 1
Urutkan berdasarkan:
1. Judul A-Z      (Bubble Sort)
2. Harga Tertinggi (Selection Sort)
3. Tahun Terlama  (Insertion Sort)
Pilihan : █
```

Gambar 4.7 Menu User Pilihan 1

```
Pilihan : 1
+-----+-----+-----+-----+-----+-----+
| Kode | Judul                | Genre  | Tahun | Harga/Hr | Status |
+-----+-----+-----+-----+-----+-----+
| F008 | A Silent Voice        | Romance | 2016  | Rp10000  | Tersedia |
| F015 | Bird Box              | Thriller | 2018  | Rp12000  | Tersedia |
| F012 | Demon Slayer: Mugen   | Action  | 2020  | Rp13000  | Tersedia |
| F007 | Five Feet Apart       | Romance  | 2019  | Rp10000  | Tersedia |
| F014 | Gone Girl              | Thriller | 2014  | Rp12000  | Tersedia |
| F002 | Hereditary             | Horror  | 2018  | Rp12000  | Tersedia |
| F019 | How To Train Your Dragon | Animation | 2014  | Rp10000  | Tersedia |
| F003 | It (2017)              | Horror  | 2017  | Rp12000  | Tersedia |
| F010 | John Wick              | Action  | 2014  | Rp13000  | Tersedia |
| F011 | Mad Max: Fury Road     | Action  | 2015  | Rp13000  | Tersedia |
| F016 | Midsommar              | Thriller | 2019  | Rp12000  | Tersedia |
| F018 | Puss in Boots 2        | Comedy  | 2022  | Rp10000  | Tersedia |
| F013 | Saltburn               | Thriller | 2023  | Rp12000  | Tersedia |
| F020 | Spirited Away          | Animation | 2001  | Rp10000  | Tersedia |
| F017 | Super Mario Bros Movie | Comedy  | 2023  | Rp10000  | Tersedia |
| F004 | Talk to Me             | Horror  | 2023  | Rp12000  | Tersedia |
| F009 | The Dark Knight        | Action  | 2008  | Rp13000  | Tersedia |
| F001 | The Nun II             | Horror  | 2023  | Rp12000  | Tersedia |
| F006 | To All The Boys        | Romance  | 2018  | Rp10000  | Tersedia |
| F005 | Your Name              | Romance  | 2016  | Rp10000  | Tersedia |
+-----+-----+-----+-----+-----+-----+

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : █
```

Gambar 4.8 User Pilih Urutan 1

Pilihan : 2

Kode	Judul	Genre	Tahun	Harga/Hr	Status
F012	Demon Slayer: Mugen	Action	2020	Rp13000	Tersedia
F010	John Wick	Action	2014	Rp13000	Tersedia
F011	Mad Max: Fury Road	Action	2015	Rp13000	Tersedia
F009	The Dark Knight	Action	2008	Rp13000	Tersedia
F014	Gone Girl	Thriller	2014	Rp12000	Tersedia
F002	Hereditary	Horror	2018	Rp12000	Tersedia
F003	It (2017)	Horror	2017	Rp12000	Tersedia
F015	Bird Box	Thriller	2018	Rp12000	Tersedia
F016	Midsommar	Thriller	2019	Rp12000	Tersedia
F013	Saltburn	Thriller	2023	Rp12000	Tersedia
F004	Talk to Me	Horror	2023	Rp12000	Tersedia
F001	The Nun II	Horror	2023	Rp12000	Tersedia
F008	A Silent Voice	Romance	2016	Rp10000	Tersedia
F020	Spirited Away	Animation	2001	Rp10000	Tersedia
F017	Super Mario Bros Movie	Comedy	2023	Rp10000	Tersedia
F019	How To Train Your Dragon	Animation	2014	Rp10000	Tersedia
F007	Five Feet Apart	Romance	2019	Rp10000	Tersedia
F018	Puss in Boots 2	Comedy	2022	Rp10000	Tersedia
F006	To All The Boys	Romance	2018	Rp10000	Tersedia
F005	Your Name	Romance	2016	Rp10000	Tersedia

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga [Binary Search]
3. Cari film by genre [Linear Search]
0. Kembali
Pilihan : █

Gambar 4.9 User Pilih Urutan 2

Pilihan : 3

Kode	Judul	Genre	Tahun	Harga/Hr	Status
F020	Spirited Away	Animation	2001	Rp10000	Tersedia
F009	The Dark Knight	Action	2008	Rp13000	Tersedia
F010	John Wick	Action	2014	Rp13000	Tersedia
F014	Gone Girl	Thriller	2014	Rp12000	Tersedia
F019	How To Train Your Dragon	Animation	2014	Rp10000	Tersedia
F011	Mad Max: Fury Road	Action	2015	Rp13000	Tersedia
F008	A Silent Voice	Romance	2016	Rp10000	Tersedia
F005	Your Name	Romance	2016	Rp10000	Tersedia
F003	It (2017)	Horror	2017	Rp12000	Tersedia
F002	Hereditary	Horror	2018	Rp12000	Tersedia
F015	Bird Box	Thriller	2018	Rp12000	Tersedia
F006	To All The Boys	Romance	2018	Rp10000	Tersedia
F016	Midsommar	Thriller	2019	Rp12000	Tersedia
F007	Five Feet Apart	Romance	2019	Rp10000	Tersedia
F012	Demon Slayer: Mugen	Action	2020	Rp13000	Tersedia
F018	Puss in Boots 2	Comedy	2022	Rp10000	Tersedia
F013	Saltburn	Thriller	2023	Rp12000	Tersedia
F004	Talk to Me	Horror	2023	Rp12000	Tersedia
F001	The Nun II	Horror	2023	Rp12000	Tersedia
F017	Super Mario Bros Movie	Comedy	2023	Rp10000	Tersedia

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga [Binary Search]
3. Cari film by genre [Linear Search]
0. Kembali
Pilihan : █

Gambar 4.10 User Pilih Urut 3

```

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : 1
Kode film : F001
+-----+
| DETAIL FILM |
+-----+
Kode       : F001
Judul      : The Nun II
Genre      : Horror
Tahun Rilis : 2023
Harga/Hari : Rp12000
Studio     : Warner Bros
Negara     : Amerika
Status     : Tersedia
+-----+
[Tekan Enter...]

```

Gambar 4.11 Hasil Pilih Aksi 1

```

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : 2
Masukkan harga yang ingin dicari (Rp) : 13000
[v] Film dengan harga Rp13000 ditemukan!
+-----+
| DETAIL FILM |
+-----+
Kode       : F010
Judul      : John Wick
Genre      : Action
Tahun Rilis : 2014
Harga/Hari : Rp13000
Studio     : Lionsgate
Negara     : Amerika
Status     : Tersedia
+-----+
Film lain dengan harga sama:
+-----+
| DETAIL FILM |
+-----+
Kode       : F009
Judul      : The Dark Knight
Genre      : Action
Tahun Rilis : 2008
Harga/Hari : Rp13000
Studio     : Warner Bros
Negara     : Amerika
Status     : Tersedia
+-----+

```

Gambar 4.12 Hasil Pilih Aksi 2

```
Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : 3
Genre yang tersedia: Horror, Romance, Action, Thriller, Comedy, Animation
Masukkan genre yang dicari : Comedy
Hasil pencarian genre "Comedy" :
+-----+-----+-----+-----+-----+
| Kode | Judul                | Genre  | Tahun | Harga/Hr | Status |
+-----+-----+-----+-----+-----+
| F018 | Puss in Boots 2      | Comedy | 2022  | Rp10000  | Tersedia |
| F017 | Super Mario Bros Movie | Comedy | 2023  | Rp10000  | Tersedia |
+-----+-----+-----+-----+-----+
[Tekan Enter...]
```

Gambar 4.13 Hasil Pilih Aksi 3

```
Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : 0
=====
MENU USER | Nurul
1. Lihat Film
2. Sewa Film
3. Kembalikan
4. Riwayat Sewa
0. Logout
-----
Pilihan : 
```

Gambar 4.14 Hasil Pilih Aksi 0

```

Pilihan : 2
=== SEWA FILM ===
+-----+-----+-----+-----+
| Kode | Judul | Genre | Harga/Hr |
+-----+-----+-----+-----+
| F001 | The Nun II | Horror | Rp12000 |
| F002 | Hereditary | Horror | Rp12000 |
| F003 | It (2017) | Horror | Rp12000 |
| F004 | Talk to Me | Horror | Rp12000 |
| F005 | Your Name | Romance | Rp10000 |
| F006 | To All The Boys | Romance | Rp10000 |
| F007 | Five Feet Apart | Romance | Rp10000 |
| F008 | A Silent Voice | Romance | Rp10000 |
| F009 | The Dark Knight | Action | Rp13000 |
| F010 | John Wick | Action | Rp13000 |
| F011 | Mad Max: Fury Road | Action | Rp13000 |
| F012 | Demon Slayer: Mugen | Action | Rp13000 |
| F013 | Saltburn | Thriller | Rp12000 |
| F014 | Gone Girl | Thriller | Rp12000 |
| F015 | Bird Box | Thriller | Rp12000 |
| F016 | Midsommar | Thriller | Rp12000 |
| F017 | Super Mario Bros Movie | Comedy | Rp10000 |
| F018 | Puss in Boots 2 | Comedy | Rp10000 |
| F019 | How To Train Your Dragon | Animation | Rp10000 |
| F020 | Spirited Away | Animation | Rp10000 |
+-----+-----+-----+-----+

Kode film (Enter -> Kembali) : F012
Tanggal Sewa:
Hari (1-31) : 4
Bulan (1-12) : 6
Tahun : 2025
Durasi (hari, min 1) : 6
Film : Demon Slayer: Mugen
Tgl Sewa : 04/06/2025
Wajib Kembali : 10/06/2025
Total Bayar : Rp78000
Denda : Rp5000/hari jika terlambat
Konfirmasi? (y/n) : y
[v] Berhasil menyewa "Demon Slayer: Mugen"!
[Tekan Enter...]

```

Gambar 4.15 Menu User Pilihan 2

```

Kode film (Enter -> Kembali) : F012
Tanggal Sewa:
Hari (1-31) : 4
Bulan (1-12) : 6
Tahun : 2025
Durasi (hari, min 1) : 6
Film : Demon Slayer: Mugen
Tgl Sewa : 04/06/2025
Wajib Kembali : 10/06/2025
Total Bayar : Rp78000
Denda : Rp5000/hari jika terlambat
Konfirmasi? (y/n) : n
Dibatalkan.

```

Gambar 4.16 Film Tidak Jadi Disewa

```

-----
Pilihan : 3
=== KEMBALIKAN FILM ===
+-----+-----+-----+
| ID | Judul                  | Tgl Sewa | Wajib Kemb |
+-----+-----+-----+
| 1  | Demon Slayer: Mugen    | 04/06/2025 | 10/06/2025 |
+-----+-----+-----+

ID sewa : 1
Tanggal Kembali:
Hari (1-31) : 9
Bulan (1-12) : 6
Tahun      : 2025
[v] Film berhasil dikembalikan!
Tepat waktu, tidak ada denda.
TOTAL BAYAR: Rp78000
[Tekan Enter...]

```

Gambar 4.17 Menu User Pilihan 3

```

Pilihan : 3
=== KEMBALIKAN FILM ===
+-----+-----+-----+
| ID | Judul                  | Tgl Sewa | Wajib Kemb |
+-----+-----+-----+
| 1  | Demon Slayer: Mugen    | 04/06/2025 | 10/06/2025 |
+-----+-----+-----+

ID sewa : 1
Tanggal Kembali:
Hari (1-31) : 11
Bulan (1-12) : 6
Tahun      : 2025
[v] Film berhasil dikembalikan!
Terlambat 1 hari
Denda      : Rp5000
TOTAL BAYAR: Rp83000
[Tekan Enter...]

```

Gambar 4.18 Jika Terlambat

```

-----
Pilihan : 4
=== RIWAYAT SEWA - Nurul ===
+-----+-----+-----+
| ID | Judul                  | Total    | Status    |
+-----+-----+-----+
| 1  | Demon Slayer: Mugen    | Rp83000  | Selesai   |
+-----+-----+-----+

[Tekan Enter...]

```

Gambar 4.19 Menu User Pilihan 4

```

=====
MENU ADMIN | Admin
1. Lihat Film
2. Sewa Film
3. Kembalikan
4. Riwayat Sewa
-----
5. Tambah Film
6. Edit Film
7. Hapus Film
8. Semua Riwayat
0. Logout
-----
Pilihan : 

```

Gambar 4.20 Menu Admin

Pilihan : 5
 === TAMBAH FILM ===
 Daftar film saat ini:

Kode	Judul	Genre	Tahun	Harga/Hr
F001	The Nun II	Horror	2023	Rp12000
F002	Hereditary	Horror	2018	Rp12000
F003	It (2017)	Horror	2017	Rp12000
F004	Talk to Me	Horror	2023	Rp12000
F005	Your Name	Romance	2016	Rp10000
F006	To All The Boys	Romance	2018	Rp10000
F007	Five Feet Apart	Romance	2019	Rp10000
F008	A Silent Voice	Romance	2016	Rp10000
F009	The Dark Knight	Action	2008	Rp13000
F010	John Wick	Action	2014	Rp13000
F011	Mad Max: Fury Road	Action	2015	Rp13000
F012	Demon Slayer: Mugen	Action	2020	Rp13000
F013	Saltburn	Thriller	2023	Rp12000
F014	Gone Girl	Thriller	2014	Rp12000
F015	Bird Box	Thriller	2018	Rp12000
F016	Midsommar	Thriller	2019	Rp12000
F017	Super Mario Bros Movie	Comedy	2023	Rp10000
F018	Puss in Boots 2	Comedy	2022	Rp10000
F019	How To Train Your Dragon	Animation	2014	Rp10000
F020	Spirited Away	Animation	2001	Rp10000

Kode : F021
 Judul : My Neighbor Totoro
 Genre : Comedy
 Tahun : 1998
 Harga : 12000
 Studio : Ghibli
 Negara : Jepang
 [v] Film ditambahkan!
 [Tekan Enter...]

Gambar 4.21 Menu Admin Pilihan 5

Pilihan : 1

Kode	Judul	Genre	Tahun	Harga/Hr	Status
F001	The Nun II	Horror	2023	Rp12000	Tersedia
F002	Hereditary	Horror	2018	Rp12000	Tersedia
F003	It (2017)	Horror	2017	Rp12000	Tersedia
F004	Talk to Me	Horror	2023	Rp12000	Tersedia
F005	Your Name	Romance	2016	Rp10000	Tersedia
F006	To All The Boys	Romance	2018	Rp10000	Tersedia
F007	Five Feet Apart	Romance	2019	Rp10000	Tersedia
F008	A Silent Voice	Romance	2016	Rp10000	Tersedia
F009	The Dark Knight	Action	2008	Rp13000	Tersedia
F010	John Wick	Action	2014	Rp13000	Tersedia
F011	Mad Max: Fury Road	Action	2015	Rp13000	Tersedia
F012	Demon Slayer: Mugen	Action	2020	Rp13000	Tersedia
F013	Saltburn	Thriller	2023	Rp12000	Tersedia
F014	Gone Girl	Thriller	2014	Rp12000	Tersedia
F015	Bird Box	Thriller	2018	Rp12000	Tersedia
F016	Midsommar	Thriller	2019	Rp12000	Tersedia
F017	Super Mario Bros Movie	Comedy	2023	Rp10000	Tersedia
F018	Puss in Boots 2	Comedy	2022	Rp10000	Tersedia
F019	How To Train Your Dragon	Animation	2014	Rp10000	Tersedia
F020	Spirited Away	Animation	2001	Rp10000	Tersedia
F021	My Neighbor Totoro	Comedy	1998	Rp12000	Tersedia

[Tekan Enter...]

Gambar 4.22 Hasil Penambahan

```

Pilihan : 6
=== EDIT FILM ===
+-----+-----+-----+-----+-----+
| Kode | Judul | Genre | Tahun | Harga/Hr |
+-----+-----+-----+-----+
| F001 | The Nun II | Horror | 2023 | Rp12000 |
| F002 | Hereditary | Horror | 2018 | Rp12000 |
| F003 | It (2017) | Horror | 2017 | Rp12000 |
| F004 | Talk to Me | Horror | 2023 | Rp12000 |
| F005 | Your Name | Romance | 2016 | Rp10000 |
| F006 | To All The Boys | Romance | 2018 | Rp10000 |
| F007 | Five Feet Apart | Romance | 2019 | Rp10000 |
| F008 | A Silent Voice | Romance | 2016 | Rp10000 |
| F009 | The Dark Knight | Action | 2008 | Rp13000 |
| F010 | John Wick | Action | 2014 | Rp13000 |
| F011 | Mad Max: Fury Road | Action | 2015 | Rp13000 |
| F012 | Demon Slayer: Mugen | Action | 2020 | Rp13000 |
| F013 | Saltburn | Thriller | 2023 | Rp12000 |
| F014 | Gone Girl | Thriller | 2014 | Rp12000 |
| F015 | Bird Box | Thriller | 2018 | Rp12000 |
| F016 | Midsommar | Thriller | 2019 | Rp12000 |
| F017 | Super Mario Bros Movie | Comedy | 2023 | Rp10000 |
| F018 | Puss in Boots 2 | Comedy | 2022 | Rp10000 |
| F019 | How To Train Your Dragon | Animation | 2014 | Rp10000 |
| F020 | Spirited Away | Animation | 2001 | Rp10000 |
| F021 | My Neighbor Totoro | Comedy | 1998 | Rp12000 |
+-----+-----+-----+-----+
Kode film (Tekan Enter -> kembali) : F021
[1]Judul [2]Genre [3]Tahun [4]Harga [5]Studio [6]Negara
Pilihan : 3
Tahun baru : 1988
[v] Film diedit!
[Tekan Enter...]==

```

Gambar 4.23 Menu Admin Pilihan 6

```

Pilihan : 1
+-----+-----+-----+-----+-----+
| Kode | Judul | Genre | Tahun | Harga/Hr | Status |
+-----+-----+-----+-----+-----+
| F001 | The Nun II | Horror | 2023 | Rp12000 | Tersedia |
| F002 | Hereditary | Horror | 2018 | Rp12000 | Tersedia |
| F003 | It (2017) | Horror | 2017 | Rp12000 | Tersedia |
| F004 | Talk to Me | Horror | 2023 | Rp12000 | Tersedia |
| F005 | Your Name | Romance | 2016 | Rp10000 | Tersedia |
| F006 | To All The Boys | Romance | 2018 | Rp10000 | Tersedia |
| F007 | Five Feet Apart | Romance | 2019 | Rp10000 | Tersedia |
| F008 | A Silent Voice | Romance | 2016 | Rp10000 | Tersedia |
| F009 | The Dark Knight | Action | 2008 | Rp13000 | Tersedia |
| F010 | John Wick | Action | 2014 | Rp13000 | Tersedia |
| F011 | Mad Max: Fury Road | Action | 2015 | Rp13000 | Tersedia |
| F012 | Demon Slayer: Mugen | Action | 2020 | Rp13000 | Tersedia |
| F013 | Saltburn | Thriller | 2023 | Rp12000 | Tersedia |
| F014 | Gone Girl | Thriller | 2014 | Rp12000 | Tersedia |
| F015 | Bird Box | Thriller | 2018 | Rp12000 | Tersedia |
| F016 | Midsommar | Thriller | 2019 | Rp12000 | Tersedia |
| F017 | Super Mario Bros Movie | Comedy | 2023 | Rp10000 | Tersedia |
| F018 | Puss in Boots 2 | Comedy | 2022 | Rp10000 | Tersedia |
| F019 | How To Train Your Dragon | Animation | 2014 | Rp10000 | Tersedia |
| F020 | Spirited Away | Animation | 2001 | Rp10000 | Tersedia |
| F021 | My Neighbor Totoro | Comedy | 1988 | Rp12000 | Tersedia |
+-----+-----+-----+-----+-----+
[Tekan Enter...]|

```

Gambar 4.24 Hasil Pengeditan


```

Pilihan : 7
=== HAPUS FILM ===
+-----+-----+-----+-----+-----+
| Kode | Judul | Genre | Tahun | Status |
+-----+-----+-----+-----+
| F001 | The Nun II | Horror | 2023 | Tersedia |
| F002 | Hereditary | Horror | 2018 | Tersedia |
| F003 | It (2017) | Horror | 2017 | Tersedia |
| F004 | Talk to Me | Horror | 2023 | Tersedia |
| F005 | Your Name | Romance | 2016 | Tersedia |
| F006 | To All The Boys | Romance | 2018 | Tersedia |
| F007 | Five Feet Apart | Romance | 2019 | Tersedia |
| F008 | A Silent Voice | Romance | 2016 | Tersedia |
| F009 | The Dark Knight | Action | 2008 | Tersedia |
| F010 | John Wick | Action | 2014 | Tersedia |
| F011 | Mad Max: Fury Road | Action | 2015 | Tersedia |
| F012 | Demon Slayer: Mugen | Action | 2020 | Tersedia |
| F013 | Saltburn | Thriller | 2023 | Tersedia |
| F014 | Gone Girl | Thriller | 2014 | Tersedia |
| F015 | Bird Box | Thriller | 2018 | Tersedia |
| F016 | Midsommar | Thriller | 2019 | Tersedia |
| F017 | Super Mario Bros Movie | Comedy | 2023 | Tersedia |
| F018 | Puss in Boots 2 | Comedy | 2022 | Tersedia |
| F019 | How To Train Your Dragon | Animation | 2014 | Tersedia |
| F020 | Spirited Away | Animation | 2001 | Tersedia |
| F021 | My Neighbor Totoro | Comedy | 1988 | Tersedia |
+-----+-----+-----+-----+
Kode film (Tekan Enter -> kembali) : F021
Hapus "My Neighbor Totoro"? (y/n) : y
[v] Film dihapus!
[Tekan Enter...]

```

Gambar 4.25 Menu Admin Pilihan 7

```

Pilihan : 1
+-----+-----+-----+-----+-----+
| Kode | Judul | Genre | Tahun | Harga/Hr | Status |
+-----+-----+-----+-----+-----+
| F001 | The Nun II | Horror | 2023 | Rp12000 | Tersedia |
| F002 | Hereditary | Horror | 2018 | Rp12000 | Tersedia |
| F003 | It (2017) | Horror | 2017 | Rp12000 | Tersedia |
| F004 | Talk to Me | Horror | 2023 | Rp12000 | Tersedia |
| F005 | Your Name | Romance | 2016 | Rp10000 | Tersedia |
| F006 | To All The Boys | Romance | 2018 | Rp10000 | Tersedia |
| F007 | Five Feet Apart | Romance | 2019 | Rp10000 | Tersedia |
| F008 | A Silent Voice | Romance | 2016 | Rp10000 | Tersedia |
| F009 | The Dark Knight | Action | 2008 | Rp13000 | Tersedia |
| F010 | John Wick | Action | 2014 | Rp13000 | Tersedia |
| F011 | Mad Max: Fury Road | Action | 2015 | Rp13000 | Tersedia |
| F012 | Demon Slayer: Mugen | Action | 2020 | Rp13000 | Tersedia |
| F013 | Saltburn | Thriller | 2023 | Rp12000 | Tersedia |
| F014 | Gone Girl | Thriller | 2014 | Rp12000 | Tersedia |
| F015 | Bird Box | Thriller | 2018 | Rp12000 | Tersedia |
| F016 | Midsommar | Thriller | 2019 | Rp12000 | Tersedia |
| F017 | Super Mario Bros Movie | Comedy | 2023 | Rp10000 | Tersedia |
| F018 | Puss in Boots 2 | Comedy | 2022 | Rp10000 | Tersedia |
| F019 | How To Train Your Dragon | Animation | 2014 | Rp10000 | Tersedia |
| F020 | Spirited Away | Animation | 2001 | Rp10000 | Tersedia |
+-----+-----+-----+-----+
[Tekan Enter...]

```

Gambar 4.26 Hasil Penghapusan

```

Pilihan : 8
=== SEMUA RIWAYAT SEWA ===
+-----+-----+-----+-----+-----+
| ID | Penyewa | Film | Total | Status |
+-----+-----+-----+-----+-----+
| 1 | Nurul | Demon Slayer: Mugen | Rp83000 | Selesai |
+-----+-----+-----+-----+
[Tekan Enter...]

```

Gambar 4.27 Menu Admin Pilihan 8

```
=====
MENU ADMIN | Admin
1. Lihat Film
2. Sewa Film
3. Kembalikan
4. Riwayat Sewa
-----
5. Tambah Film
6. Edit Film
7. Hapus Film
8. Semua Riwayat
0. Logout
-----
Pilihan : 0
Sampai jumpa, Admin!
[Tekan Enter...]
```

Gambar 4.28 Menu User dan Admin Pilihan 0

5. Langkah-langkah GIT

5.1 GIT Add

```
PS C:\Users\LOQ\Documents\praktikum-apl> git add .
```

Gambar 5.1 Git Add

Digunakan untuk menambahkan perubahan file ke staging area (area persiapan).

5.2 GIT Commit

```
PS C:\Users\LOQ\Documents\praktikum-apl> git commit -m "Codingan Post test 7"
[main a9031e3] Codingan Post test 7
6 files changed, 1438 insertions(+)
create mode 100644 A2'25/Pertemuan-7/Pertemuan-7.cpp
create mode 100644 A2'25/Pertemuan-7/Pertemuan-7.exe
create mode 100644 A2'25/Pertemuan-7/pertemuan_7.h
create mode 100644 post-test/post-test-apl-7/2509106033_NurulLatida_PT-7.cpp
create mode 100644 post-test/post-test-apl-7/2509106033_NurulLatida_PT-7.exe
create mode 100644 post-test/post-test-apl-7/validasi.h
```

Gambar 5.2 Git Commit

Digunakan untuk menyimpan perubahan yang sudah di-add ke dalam repository lokal dengan pesan tertentu.

5.3 GIT Push

```
PS C:\Users\LOQ\Documents\praktikum-apl> git push
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 20 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 135.01 KiB | 5.19 MiB/s, done.
Total 12 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:  https://github.com/nulaatifa/praktikum-apl.git
To https://github.com/nulaatifa/Praktikum_Algoritma_Pemograman_Lanjut.git
9ed8b8e..a9031e3  main -> main
PS C:\Users\LOQ\Documents\praktikum-apl>
```

Gambar 5.3 Git Push

Digunakan untuk mengirim hasil commit dari repository lokal ke repository remote (misalnya ke GitHub).